

SENIOR Geography 2

GLOBAL ISSUES

Kate Lanceley & Bruce Pacey



Quality ready-to-use
resources



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INTRODUCTION

Instant Lessons in Senior Geography: Book 2 – Global Issues is part of a two-book series, which is a collection of student-friendly worksheets created for Years 9–12 Geography students throughout Australia.

The worksheets explore a range of different global environmental, social and political geographical issues. They are to be used in conjunction with set texts and classroom studies. They can be used in a variety of ways, either to introduce a topic, as a review of issues already studied, to further expand students' knowledge and skills, as a summary or a replacement lesson when a teacher is away.

Emphasis has been placed on skills development. Students are required to interpret stimulus material using geographical tools. A variety of different learning activities are included in each worksheet to engage students' interest.

Some worksheets provide space for student answers, but often a separate sheet or workbook is required. Blank outline maps of the World and Asia are provided on the inside front and back covers to be used in topics where this is noted.

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1 ENERGY RESOURCES

Introduction As the world economy continues to grow, newly emerging prosperity in some of the most populous nations, such as China, will add substantially to the demand for energy. Pressure on the existing energy sources will increase, as will the need to expand alternative supplies.

Classifying energy sources

Resources may be classified as **renewable** or **non-renewable**. Renewable resources may be further classified as perpetual, being in continuous supply despite human use, or manageable, which only remain renewable with appropriate human management.

1 Classify each of the following energy resources using one of these three codes:

- N non-renewable
- RP renewable perpetual
- RM renewable with management

Definitions **renewable** – those resources which due to continuous natural supply, recycling or reproduction continue to be available despite human use. For some resources, this depends on the rate of human use being slower than the rate of natural replacement.

non-renewable – those resources which are in fixed, limited or finite in supply and once used are not replaced by natural processes (except after very long periods of geologic time). Some of these may be capable of recycling after human use to extend the period of their use.

Energy source	Classification	Energy source	Classification
1 biomass		9 peat	
2 coal		10 shale oil	
3 crude oil		11 solar	
4 geothermal		12 tidal	
5 hydro power		13 uranium	
6 marine currents		14 wave	
7 natural gas		15 wind	
8 ocean thermal		16 wood	

Patterns of energy use

Primary energy sources, 2000

Energy source	Global use %
Crude oil	35
Coal	22
Natural gas	21
Nuclear	7
Hydro-electricity	2
Traditional biofuels	11
Other renewables	2

[Source : United nations]

2a Construct a divided bar graph next to the table above showing the sources of primary energy in 2000.

2b Calculate the percentages of the 2000 energy that are obtained from renewable sources and non-renewable sources.

2c What are ‘traditional biofuels’? List as many types of these fuels as you can.

2d Who would depend heavily on these types of fuels? What problems would these people encounter in their use of, and dependence on these types of fuels?

2e What energy sources would be in the ‘Other renewables’ category? Suggest why the level of use of these fuels is so low and their uptake has been so slow.

Changing patterns of energy use

Method for calculating relative or percentage change over time:
First obtain the absolute change by subtracting the initial figure from the later one. Then divide the absolute change by the initial figure, and multiply by 100%.

	1950	2000	Percentage change
World energy use (energy units per year)	100	429	
World population (billions)	2.7	6.0	
Energy use per billion people			

[Data source: The World Resources Institute, 2004]

3a In the table above calculate the percentage increase in world population and world energy use between 1950 and 2000, and the energy use per billion people in 1950 and 2000. Write a brief comment on the results of your calculations.

3b Write a paragraph commenting on the results of your calculations. What do they suggest about energy demand in the future?

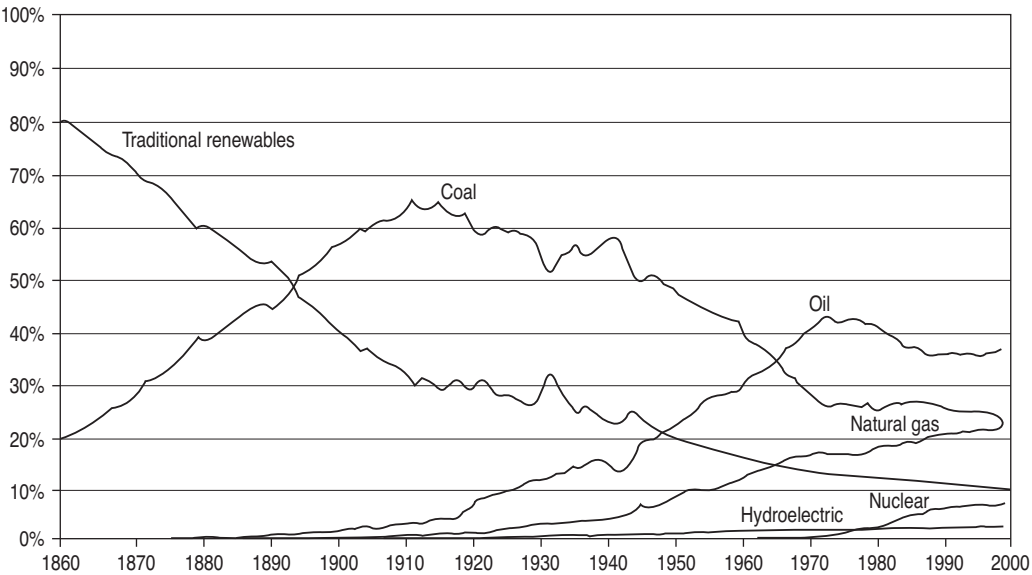
2 THE OIL CRISIS

Introduction The world's high dependence on crude oil as a major source of energy grew rapidly during the last century. Economic development, industrialisation and the technology of the internal combustion engine provided a growing market for the plentiful and growing supplies of crude. In the current century, it is expected that oil prices will rise rapidly as growing demand exceeds the dwindling supplies, and alternative fuels will become more important.

Changing energy sources

Percentage of total calculated consumption contributed by each energy source

[Source: <http://dieoff.org/page199.htm>]



1a Use the graph above to complete the table on changes in energy use, then answer the remaining questions in this section on a separate sheet of paper.

Energy source	% share 1900	% share 1950	% share 2000
Traditional renewables			
Coal			
Oil			
Natural gas			
Hydro-electricity			
Nuclear			

1b Describe the pattern of change in energy use that has occurred since 1970.

1c Explain how the oil crises of the 1970's, when OPEC restricted oil supplies to force up crude oil prices, may have influenced the post 1970 pattern.

1d The use of oil has been increasing rapidly since the 1970s. Explain how this is possible despite the trend shown in the graph.

1e Identify types of energy that are not included in the graph. Suggest why.

1f What types of fuels would be in the 'traditional renewables' category?

The dominance of oil

2a On a separate sheet construct a sector graph (pie diagram) showing the pattern of energy use from the data provided in the table below.

2b Describe the pattern.

Primary energy sources, 2000

Energy source	Usage %
Crude oil	35
Coal	22
Natural gas	21
Nuclear	7
Hydro power	2
Traditional biofuels	11
Other renewables	2

[Source: United Nations]

Using oil resources

3 On a separate sheet construct a mind map showing the different ways that energy derived from oil can be used. The following data will assist you in this task.

Global oil consumption by sector, 2000

Industry sector	Usage %
Road transport	39.0
Petro-chemical industries	9.7
Power (electricity) generation	7.8
Other industries	6.8
Domestic use	6.4
Air transport	5.9
Oil refinery use	5.8
Other uses	18.6

[Source: The Financial Times]



Replacing oil-based energy

4 Fill in or construct a table like the one below, using the example provided as a guide.

Uses of oil	Possible replacement	Advantage and/or disadvantage
Road vehicles	1 Fuel cells using hydrogen	produce no pollution, just water high cost, high energy for hydrogen production, limited refuelling sites
	2 Electric motors	
Other examples		

How long will the oil last?

With the demand for oil growing faster than new discoveries, our proven reserves of oil are expected to run out this century. There have been no new major discoveries to match the existing reserves in the last 30 years, and possible new oil fields are in remote and inaccessible or environmentally sensitive locations.

5 Examine the table at right then answer the questions. (Note: the **reserve to production ratio** in column 4 indicates how long the remaining reserves will last if production continued at the same rate as in 2003.)

a Calculate the percentage share of oil reserves for each region and add them to the table.

b In what year will the world run out of oil if there are no new discoveries made?

c Rank each region in the order from first to last in terms of when its oil supplies will run out.

- 1 _____
- 2 _____
- 3 _____
- 4 _____
- 5 _____
- 6 _____

d Write an extended response describing the global oil industry in 20 years time, assuming there are no new substantial discoveries of oil. Consider aspects such as which countries or regions will be producing and consuming the oil, what its price might be and for what purposes it will mainly be used.

Current oil reserves

Region	Proven oil reserves at end of 2003 (thousand million barrels)	% share of world total	Reserve to production ratio (years)
North America	63.6		12.2
Central & South America	102.2		41.5
Europe and Eurasia	105.9		17.1
Middle East	726.6		88.1
Africa	101.8		33.2
Asia Pacific	47.7		16.6
WORLD	1147.8	100	41

[Source: BP Statistical Review of World Energy 2004]

3 WATER RESOURCES

Introduction Freshwater is a universal necessity. It sustains all forms of life, allowing ecosystems to operate by providing food, recycling energy and removing waste. For humans, water also promotes development. Without adequate supplies, living standards decline, people suffer and economic growth declines. To focus the world's attention on current and growing water-related issues, the UN declared 2003 the International Year of Freshwater.

Water vocabulary

1 Match each of the following terms with the correct definition in the table.

aquifer, non-renewable water, renewable water, sanitation, water withdrawal, water scarcity, water stress

Definition	Term
1 Removal of freshwater for human use from any natural source or reservoir such as a lake, river or aquifer	
2 Freshwater that is regularly replenished by the hydrologic cycle for withdrawal within reasonable time limits.	
3 A layer or section of earth or rock that contains freshwater (groundwater)	
4 A situation where a country has an annual per capita water supply of 1000 to 1700 cubic metres and will experience temporary or limited water shortage	
5 Water in aquifers or other natural sources that is not recharged or recharges so slowly by natural processes that its withdrawal by humans causes depletion	
6 A situation where a country has an annual per capita water supply of less than 1000 cubic metres and will experience chronic and widespread supply problems that hinder development	
7 A term for water/systems that reduce health risk to humans, such as sewage disposal and drainage	

Water supply

Of the world's estimated 1.4 billion cubic kilometres of water, 97.5% is salt water. The remaining 2.5 % is composed mainly of:

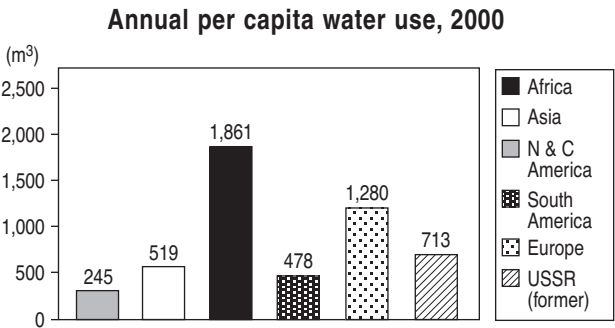
- glaciers and ice caps (68.9%)
- groundwater (30.8%)
- surface water – lakes and rivers (0.3%).

There are minute proportions stored in the atmosphere and biosphere.

2 On a separate sheet write a brief evaluation on the usefulness and ease of access for humans to each of the natural storages of water listed above. Include modifications to the storage and/or environment that would make them more useful or accessible.

Global inequalities

The amount (and quality) of water available for human use varies markedly due to natural availability from the water cycle, population levels and availability of technology to protect, treat and transport the available water.



[Source: USSR National Academy of Science]

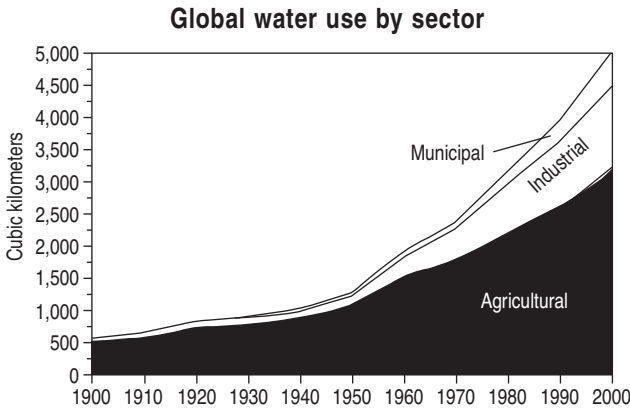
- 3a Construct a similar graph to the one on the previous page for the countries listed in the table below.
- 3b Suggest reasons for differences in the levels shown, and compare them with regional levels shown in the graph above.

Estimated per capita freshwater withdrawal, 2000

Country	Withdrawal (cubic metres per annum)
Australia	786
Egypt	809
USA	1688
China	360
France	591
Namibia	144

[Source: World Water Organisation]

Patterns of water use



- 4a Refer to the graph above. Calculate the percentages of global water withdrawal for agricultural, industrial and municipal (public and domestic) purposes in the year 2000.
- 4b In a small group, brainstorm a list of all the possible uses for fresh water. Share your list with other groups. Then decide how you might group all these uses of water into general categories such as those shown in the graph above. Develop sub-categories and show your results in a tree diagram.
- 4c Write a paragraph describing the changing pattern of water use during the 20th century. Include both total use and its composition. Suggest reasons for the patterns described.

Using our water supplies

There is considerable variation in how water is used in different places around the world, as shown in the following examples:

Country	Total fresh-water with-drawal km ³ /yr	Domestic and public use %	Industrial use %	Agricul-tural use %
Egypt	55	6	8	86
USA	469	12	46	42
Australia	15	65	2	33
Germany	59	14	68	18

[Source: World Water Organization]

- 5a Construct sector (pie) graphs showing water use for the countries shown in the table above.

For percentage data, multiply each figure by 3.6 to obtain the required angle for the graph. When comparing multiple graphs with similar data categories, it is helpful to maintain the same data order in all graphs, for example, domestic, industrial, agricultural.

- 5b Use the data in the table to describe the pattern of water use in the countries listed. Consider both total withdrawal and use by sector. Suggest reasons for the patterns described.

The future

About 40% of the world's population currently live in areas of water stress. It is estimated that this will rise to about two thirds by 2025 (5.5 billion people). Water use increased six-fold during the last century, more than twice the rate of population growth. Wetlands are being degraded, water supplies polluted and ground water is being depleted. More than 2 million people die each year from diseases related to contaminated drinking water and poor sanitation. In 1999 the world supported 6 billion people, which will grow to over 8 billion by 2025. [Source: UN 2003]

- 6 Assume you are the United Nations officer responsible for World Water Policy. Compile a list of measures that could be taken to help solve or improve the world's current and future water problems.

4 GLOBAL PATTERNS IN FISHING

Introduction Fish and seafood products have long been an important source of food for human populations. Increasing demand from population growth and alternative uses, combined with degradation of fish habitats is placing this resource under increasing stress.

Global patterns

Fish, crustaceans and molluscs provide approximately 16% of the human population’s animal protein intake.

1a Make a list of the different sources of animal protein that are included in the human diet.

1b Calculate the following and add them to the table below:

- the world total fish food supply
- the percentage share for each region
- the world average per capita fish supply.

Total and per capita fish food supply, 2000

Region	Total fish food supply (million metric tonnes)	% share	Per capita fish food supply (kg/pa)
Africa	6.2		8.0
North & Central America	8.1		16.8
South America	2.9		8.5
China	31.2		25.1
Other Asia	32.5		13.7
Europe	13.9		19.1
Oceania	0.7		22.5
WORLD		100	Average

[Source: Food and Agricultural Organization (FAO), 2000]

1c Explain why you think China is listed separately from the other Asian nations.

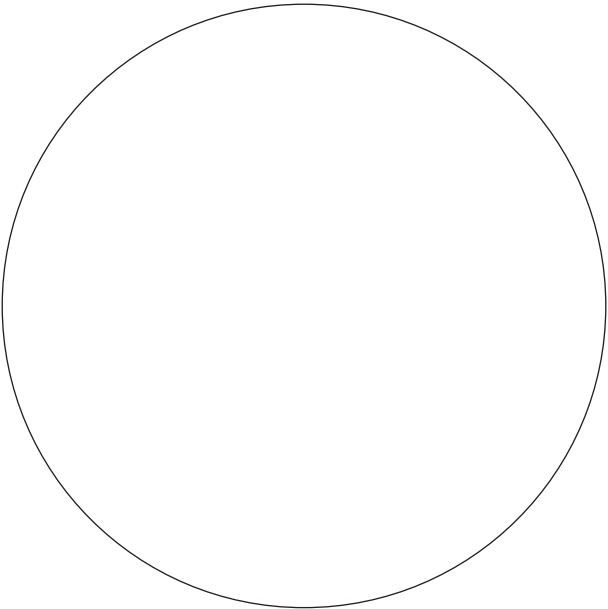
1d Explain why Oceania might have the lowest total supply, but a very high per capita supply compared to most other regions.

How fish products are used

Approximately 29% of the world’s fish catch is used for purposes other than direct human consumption (for example as stock feed, oil, and other products).

- Fish is a highly perishable product. Of the remaining 71%
- 53.7% is consumed fresh
 - 25.7% is frozen
 - 11% is canned
 - 9.6% is cured (smoked).

2 Construct a sector (pie) graph to show the use of fish from the figures above.



Supplementing the supply

3a List the advantages of sourcing more of our demand for fish from aquaculture, also known as fish farming. Are there any disadvantages or problems?

3b Use this data to draw on a separate sheet a multiple line graph showing the pattern of fish capture (traditional fishing) and aquaculture.

World fish output, metric tonnes

	1996	1997	1998	1999	2000	2001
Total capture (inland and marine)	93.5	93.9	87.3	93.2	94.8	91.3
Total aqua-culture (inland and marine)	26.7	28.6	30.5	33.4	35.6	37.5

[Source: FAO, 2002]

3c What is meant by the terms ‘inland’ fishing, and ‘marine fishing’? Give examples of locations around the world where each might occur for both capture and aquaculture.

3d Comment on the trends in your graph. Consider features such as the changing share between each source, the pattern in each over time, and suggest reasons for the patterns observed. What does it suggest about the future of the global fishing industry?

Caring for the catch – impacts on fishing

A continuing supply of fish harvested from the world’s oceans and inland waterways will depend on management of these environments, natural ecosystem change and regulation of the fishing industry. Of the world’s 15 major fishing regions, the UN states that 4 are already depleted and another 9 are declining.

4 Match each of the following actions with the main effect they would have on the state of the world’s fisheries:

overfishing, water pollution, use of driftnets, catch quotas, dam building on coastal rivers, wetland depletion, bottom trawling

Action	Effect
	1 Reduces fish stocks due to loss of the nursery areas that protect young fish on shallow coastlines and estuaries.
	2 Catching of unwanted fish species and other marine creatures (by-catch) leading to reduced ocean biodiversity.
	3 Destruction of fish habitat, and therefore food sources and protection areas for fish in shallow fishing grounds.
	4 Reduces the stock of fish available for breeding, leading to a species becoming threatened, endangered or extinct.
	5 Causes death, disease, abnormalities and reduced ability to reproduce in fish populations.
	6 Interrupts the breeding cycles for ocean species such as salmon that spawn in fresh water.
	7 A measure designed to prevent or reduce overfishing that has limited success due to global cooperation and enforcement difficulties.

5 TROPICAL RAINFORESTS AND DEFORESTATION

Introduction Tropical rainforests are found in the warm wet climates of the tropics. They are vitally important, not only for their biodiversity – over half the world’s plant and animal species live in rainforests – but for the role they play in regulating global climate, as a source of plants to produce medicines, in providing homes for indigenous people and in acting as watersheds. Despite their value the world’s tropical rainforests have experienced, and are continuing to experience, drastic rates of deforestation.

Definitions

1 Match each of the following terms by joining it to the correct definition.

a	variety of plant and animal life	watershed
b	facilities such as road and rail networks, water and power supplies	tropical rainforest
c	a community of interacting plants and animals that are interdependent	leaching
d	the downward movement of nutrients through the soil profile, which leads to the top layer of soil becoming infertile	biodiversity
e	dense forest found in the tropical areas with high humidity and heavy rainfall	infrastructure
f	the boundary between two catchment areas	ecosystem

Rainforests

2a In the table below label the following statements about rainforests true or false. Rewrite the incorrect statements correctly in the last column.

Rainforest fact or fiction?	T or F?	Correct statement
1 Forty per cent of the world’s plant and animal species live in rainforests.		
2 Australia has the largest area of untouched tropical rainforest in the world.		
3 Rainforest plants have been used to make medicines used in the treatment of cancer.		
4 When rainforest trees are cut down they release oxygen into the atmosphere, contributing to global warming.		
5 Poverty is a major cause of the destruction of rainforests.		
6 Greater use of renewable energy would help save the rainforests from destruction.		

Climate of the rainforest

Climatic data for Pangkalanbuun in Kalimantan, Indonesia

Elevation 25 metres, latitude 2° 42 S, longitude 111° 42 E

Average temperature

	Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
°C	27	27	27	27	27	28	27	27	27	27	28	27	27

Average precipitation

	Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
mm	2865	257	254	280	274	287	181	173	160	178	191	297	333

3a Use the climatic statistics for Pangkalanbuun to construct a climatic graph on a separate sheet, then answer the questions b to f under the graph. (Note: questions c to f are on the next page.)

3b Describe the climate of Pangkalanbuun.

- 3c Calculate the temperature range for Pangkalanbuun.
- 3d Explain why it has such a low temperature range.
- 3e Describe the rainfall distribution in Pangkalanbuun.
- 3f What might happen to the rainfall in Pangkalanbuun if huge amounts of rainforest were removed from the island of Borneo? Explain.

The rainforest ecosystem

In a rainforest ecosystem, there is a complex relationship between plants and animals, which are all interdependent.

4a Match each of the following terms with the correct definition in the table:

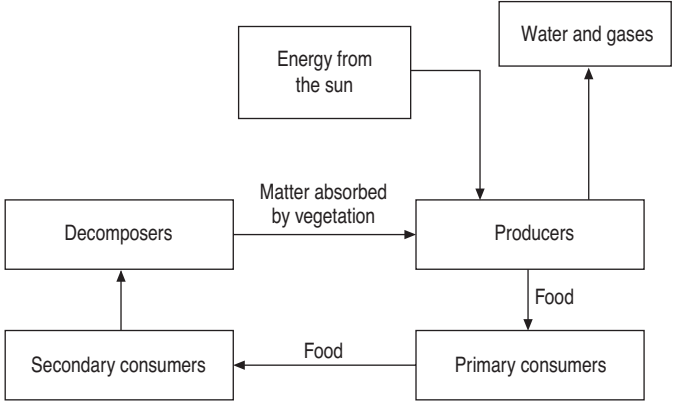
carnivores, epiphytes, herbivores, humus, leaf litter, lianas, parasites, saprophytes

1 Fungi such as mushrooms and toadstools that live on dead organic matter found on the forest floor.	
2 Plants that grow on other plants for support, eg ferns and orchids.	
3 Plants that climb upon a host tree and live off their sap.	
4 Leaves from trees and plants that have fallen to the forest floor and are decomposing to form forest soil rich in minerals and nutrients.	
5 The rich top layer of the soil made of decomposed plant or animal matter.	
6 Rope-like vines that twist themselves around rainforest trees in search of sunlight.	
7 Plant-eating animals	
8 Meat-eating animals	

4b On a separate sheet copy the diagram at top right of the rainforest ecosystem. Either draw or list examples of species that make up the rainforest ecosystem. Don't forget that there are both plant and animal species, and even some marine species.

Sustainable management

- 6 Present an argument for better managing the rainforest. Use these headings to structure your argument.
- Maintenance of genetic diversity
 - Utility value
 - Need to allow natural change to proceed
 - Intrinsic values
 - Heritage values



Causes and impacts

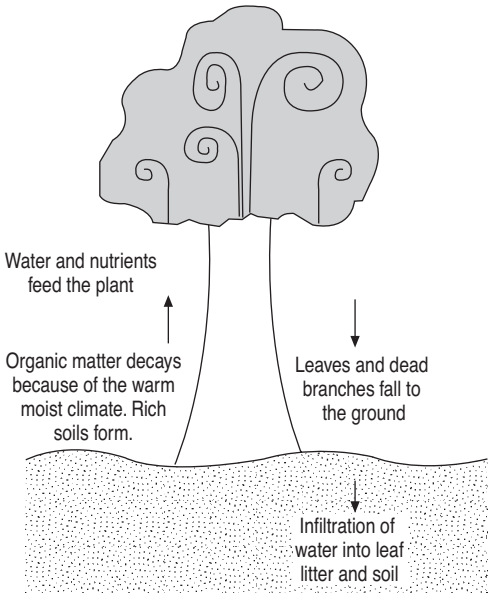
5a Draw up a table like the one below and categorise the following factors as either **causes** or **impacts** of deforestation. Add any other causes or impacts of deforestation to your table that you can think of.

cattle ranching and livestock grazing, commercial agriculture, destruction of traditional lifestyle, destruction of waterways by high sedimentation loads, drought and desertification, fuelwood collection, global warming, infrastructure development, logging, loss of biodiversity, mining and oil exploration, slash and burn farming, soil leaching, soil degradation, transmigration

Deforestation

Causes	Impacts

5b The diagram below shows the functioning of the rainforest ecosystem prior to deforestation. Copy this, then next to it draw a diagram to explain what happens to the rainforest ecosystem when the trees are removed.



6 DESERTIFICATION

Introduction Drylands are the areas bordering the world's deserts and therefore at risk from desertification. They are inhabited by more than 2 billion people, or approximately 40% of the world's population. Many of the world's poorest people, as measured by the HDI (Human Development Index) reside in these zones, making their management an even more critical and difficult task.

Desertification vocabulary

1 Match the terms provided with their correct meaning in the table below.

afforestation, biodiversity, desertification, drought, ecological refugees, evapotranspiration, fuelwood, overgrazing, salinity, windbreaks

Term	Meaning	Term	Meaning
	1 a prolonged period of below average precipitation		6 the removal of water vapour from water surfaces, soil and vegetation
	2 when plants are exposed to grazing for too long, without sufficient time to recover or regenerate, therefore leading to desertification		7 the concentration of soluble minerals in water-bodies and soil, often making them less productive or useless
	3 people who leave their land and relocate due to degradation of the environment and its inability to continue supporting them		8 a measure of the range of plant and animal life that occupies an area
	4 the act of establishing forests on land that did not previously support them		9 timber collected for use as an energy supply, especially cooking
	5 the degradation of land in arid, semi-arid and sub-humid areas resulting from various natural and/or human processes, which makes it arid and less productive		10 the planting (usually in rows) of trees and shrubs to reduce erosion and/or provide shelter for animals

The global pattern

Areas and populations affected by desertification

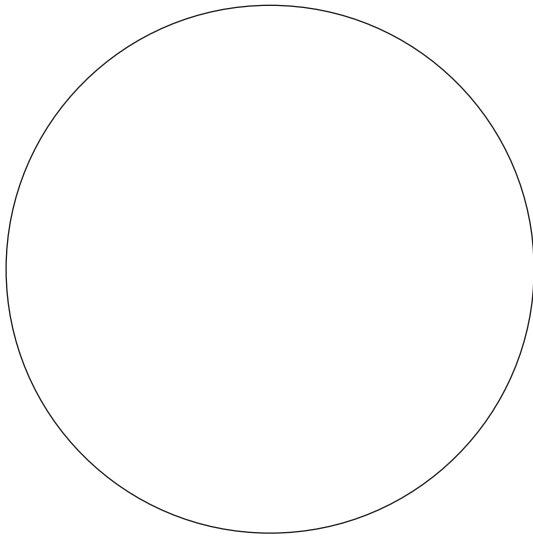
Region	Area ('000 km ²)	%	Rank	Population (millions)	%	Rank
Africa	7409			108.00		
Asia	7480			123.00		
Australia	1123			0.23		
Europe	296			16.50		
North America	2080			4.50		
South America	1620			29.00		
WORLD						

[Source: UNEP]

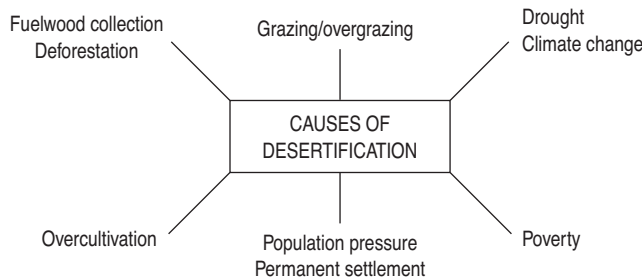
2a In the table above add the areas and populations affected to fill in the world totals.

2b Rank the regions from most affected (1) to least affected (6) for both the area and population.

- 2c** Examine your two rankings. Then develop an overall ranking based on both. Write a paragraph response in which you explain and justify the overall ranking you have reached. What were your criteria?
- 2d** Construct a sector graph (pie chart) to show the regional distribution of land affected by desertification.



Causes of desertification



- 3a** Divide into groups and brainstorm how each of these factors could affect desertification. Each group should present a short oral report to the class.
- 3b** After the talks, create a flow or sequence diagram, showing how these causes could be linked, or could follow in a sequence.

Taking remedial action

From the Rio Earth Summit, held in June 1992, the UNCCD (United Nations Convention to Combat Desertification) evolved. By August 2003 it had 190 countries as signatories. Its aim is to complement the environment and development work of the UN by specifically addressing the global desertification issue. It administers funding, research, implementation and evaluation for the fight against desertification.

- 4** Some measures used are listed below. Explain how they might help reduce the problems of desertification. Some may have multiple benefits. You may divide into groups and allocate a number of measures to each group.

1. Using remote sensing (satellite imagery)
 2. Computer simulation data
 3. Planting leguminous crops (nitrogen fixers)
 4. Placing stones around the base of trees
 5. Planting rows of shrubs or trees
 6. Spraying sand dunes with oil containing plant seeds (Middle East)
 7. Planting groups of woody trees and shrubs close to villages
 8. Bringing in hay for stock to consume
 9. Providing local people with simple solar ovens
 10. Digging shallow channels along the contours of hill slopes
 11. Laying out 1 metre square straw grids, planting a tree or shrub seedling in each
 12. Diversify agriculture / introduce additional crops
 13. Developing non agricultural industries
 14. Using surplus agricultural labour for agricultural support projects
 15. Converting land from grazing to cropping
 16. Introducing locally suited crop rotation
 17. Relocating communities
 18. Establishing efficient water storage and irrigation systems
 19. Involving local communities in preventative and restoration projects
 20. Limiting or banning the use of recreational vehicles (4WDs) in fragile areas (USA)

7 POPULATION CHANGE

Introduction The traditional ‘demographic transition’ model of population considered only birth and death rates. Initially, populations were kept small by high birth and high death rates, then falling death rates led to rapid population growth. Birth rates then began to fall until a final stage of low growth or stability was reached. Countries were seen to pass through these stages with increasing economic development. This unit explores some of the factors influencing population change and uses recent United Nations data to compare what is happening in the more and less developed regions of the world.

Basic terms

1 Match each of the terms provided with the meanings in the table below.

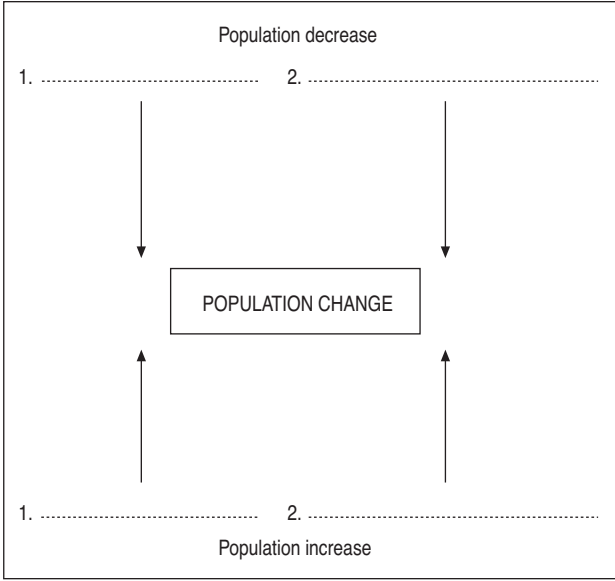
fertility, density, net migration, demography, distribution, median age, natural increase

Term	Meaning
	1 The study of human populations with emphasis on their statistical features
	2 The middle age in a population, such that there are equal numbers of people above and below that age
	3 The number of children born, on average, to women in a population during their lifetime
	4 Crude birthrate minus crude deathrate
	5 The population of a country or region, divided by the area of land it covers
	6 The pattern formed by the location of people on areas of the earth's surface
	7 The number of immigrants to an area or country less the number of emigrants in a given period of time

Change factors

2 Complete the following table with the major factors that can cause the population of a country or region to change in size over time.

Migration can make a significant difference to the population of a country or region over time. People are said to be influenced in their movements by push factors and pull factors. Complete the table below in which an example has been given of push and pull factors, and where possible give an example of where/when these may have applied in recent migration patterns.



Factor	Type	Example
1 unemployment	push	large numbers of unemployed have left the Philippines to become guest workers in Europe, North America and the Middle East
2		
Extend table to 10 examples		

3a Examine the tables below, which were sourced from the Population Division of the Department of Economic and Social Affairs of the United Nations Secretariat, World Population Prospects: The 2002 Revision and World Urbanization Prospects: The 2001 Revision, <http://esa.un.org/unpp>, 2 August 2004.

Answer true or false to the statements in the table that follows them. Use the data and your own knowledge.

More developed regions

(Europe, North America, Australia, New Zealand, Japan)

**Population change per year (thousands)
Medium variant, 2000–50**

Period	Population change per year
2000–50	2989
2005–10	2408
2010–15	1929
2015–20	1380
2020–25	796
2025–30	180
2030–35	–412
2035–40	–967
2040–45	–1412
2045–50	–1732

Less developed regions

(Africa, Asia—except Japan, Latin America, Pacific Island and Caribbean)

**Population change per year (thousands)
Medium variant, 2000–50**

Period	Population change per year
2000–05	73 620
2005–10	72 923
2010–15	71 464
2015–20	67 218
2020–25	61 448
2025–30	55 558
2030–35	50 019
2035–40	44 049
2040–45	37 573
2045–50	30 598

[Source: World Population Prospects: The 2002 Revision and World Urbanization Prospects: The 2001 Revision, UN, 2004]

Statement about population change	True or false?
1 More developed regions will experience population fall while less developed regions will experience population growth for the whole of period shown	
2 The total amount of population change is greater in less developed countries than in more developed countries	
3 More developed countries will have an ageing population, and less developed countries a stable age structure during the period shown.	
4 More develop countries usually experience slower population growth due to falling birthrates	
5 Most migration is usually from more developed regions to less developed regions	

3b From the tables, identify one similarity and two differences in the patterns of population change for the more and less developed regions over the period shown.

Similarity _____ Differences _____

4 Complete the following table giving advantages and disadvantages for a country experiencing population change from different causes.

Change	Advantages	Disadvantages
1 increased birthrate		
2 decreased birthrate		
3 increased immigration		
4 increased emigration		

8 GLOBAL POPULATION GROWTH

Introduction According to the United Nations, the world's population is expected to increase by 2.6 billion from 6.3 billion in 2003 to 8.9 billion in 2050. Currently the growth rate of the world's population is 1.2 percent. About 77 million people are being added to the world's population each year.

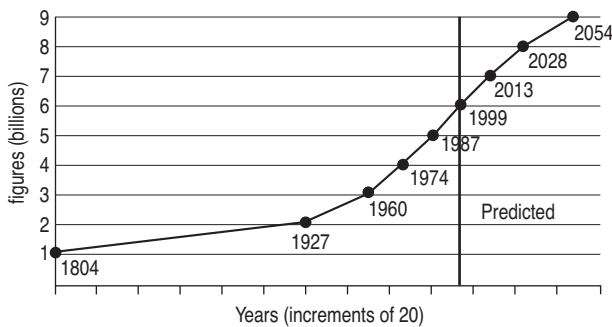
Definitions

1 Match each of the following definitions by joining it to the correct term.

a	the difference between two values divided by the original number times 100	percentage increase
b	the average number of children per woman	natural increase
c	share divided by the total number times 100	life expectancy
d	where things are	percentage share
e	the number of years that a person can expect to live	fertility rate
f	the increase of population by subtracting the number of deaths from the number of births	infant mortality rate
g	the number of deaths of infants in the first year of life per 1000 live births	spatial distribution

Population growth

World population growth 1804–2054



[Source: 2003 World Population Data Sheet of the Population Reference Bureau]

2a Using statistics compare the growth in the world's population in the nineteenth and twentieth centuries.

2b Suggest reasons for the differences in global population growth during each of these periods.

2c Compare the length of time it took the world's population to grow from 1 to 4 billion with the length of time it is expected to take the world's population to grow from 6 to 9 billion.

2d Suggest reasons for this difference.

3a Refer to the table at right and calculate the percentage share of the world's population of each of the More Developed Countries and the Less Developed Countries.

3b Using the figures for rate of natural increase compare the number of people being added to the population of the More Developed Countries and the number of people being added to the Less Developed Countries each year.

3c Compare the fertility rate in the less developed and more developed countries. Suggest reasons for the difference in fertility rate in each of these two groups of countries.

Population characteristics of developed and developing countries

Characteristic	More developed	Less developed
Population 2003 (millions)	1202	5112
Birthrate per 1000 pop	11	24
Deathrate per 1000 pop	10	8
Rate of natural increase %	0.1	1.6
Projected population 2025 (millions)	1260	6647
Projected population 2050 (millions)	1257	7940
Infant mortality rate	7	61
Total fertility rate	1.5	3.1
% population aged <15	18	33
% population aged 65+	15	5
Life expectancy at birth (years)		
Total	76	65
Male	72	63
Female	79	66
% urban	75	40

[Source: 2003 World Population Data Sheet of the Population Reference Bureau]

3d Using the figures for percentage of the population <15 and percentage >65, compare the age structure of the More Developed and the Less Developed Countries. Outline the challenges that the age structure of each of these groups of countries is likely to present.

3e Explain why there is such a wide disparity in life expectancy between the country types for the More Developed and the Less Developed Countries.

3f Suggest reasons for the huge difference in the proportion of people living in urban areas between the More Developed and the Less Developed Countries.

Areas of growth, slower growth and decline

Major growth areas 2003–50

- Middle Africa is expected to be the fastest growing region in the first half of the 21st century, increasing 193 percent
- Western Africa is expected to increase by 142 percent
- In Latin America and the Caribbean, Central America is expected to grow by 60 percent
- The Caribbean is expected to grow by 36 percent
- South America is expected to grow by 42 percent
- Western Asia is expected to grow by 105 percent

Slower growing and declining areas 2003–50

- In Southern Africa where the population has been decimated by HIV/AIDs the population is expected to decline by 22 percent
- Eastern Asia, home to 1.5 billion people is expected to increase by just 5 percent
- Northern Europe is expected to grow slightly, with a 6 percent gain, while the rest of the continent will decline.

4a Describe the spatial distribution of the regions that are expected to experience substantial population growth in the first half of the twenty first century.

4b Suggest reasons for why these regions are experiencing substantial population growth.

4c Describe the spatial distribution of the regions that are expected to experience substantial decline or slower growth in the first half of the twenty first century.

4d Suggest reasons for why these regions are experiencing decline or slower population growth.

Largest countries

World's largest countries in 2003

Rank	Country	Population (millions)
1	China	1289
2	India	1069
3	USA	292
4	Indonesia	220
5	Brazil	176
6	Pakistan	149
7	Bangladesh	147
8	Russia	146
9	Nigeria	134
10	Japan	128

[Source: 2003 World Population Data Sheet, Population Reference Bureau]

World's largest countries in 2050 (projected)

Rank	Country	Population (millions)
1	India	1628
2	China	1394
3	USA	422
4	Pakistan	349
5	Indonesia	316
6	Nigeria	307
7	Bangladesh	255
8	Brazil	221
9	Dem Rep of Congo	181
10	Ethiopia	173

[Source: 2003 World Population Data Sheet, Population Reference Bureau]

5a Which countries listed in the top 10 countries in terms of population 2003 are not expected to be in the top 10 countries in 2050? Suggest possible reasons for this.

5b Calculate the expected percentage increase in the populations of each of the following countries between 2003 and 2050:

- China
- India

5c Suggest possible reasons for the difference in the growth of China and India.

5d Describe the spatial distribution of the expected world's 10 largest countries in 2050.

9 AGEING OF THE WORLD'S POPULATION

Introduction The ageing of the world's population will have a range of economic and social implications for both developed and developing nations.

Definitions

1 Match each of the following definitions by joining it to the correct term.

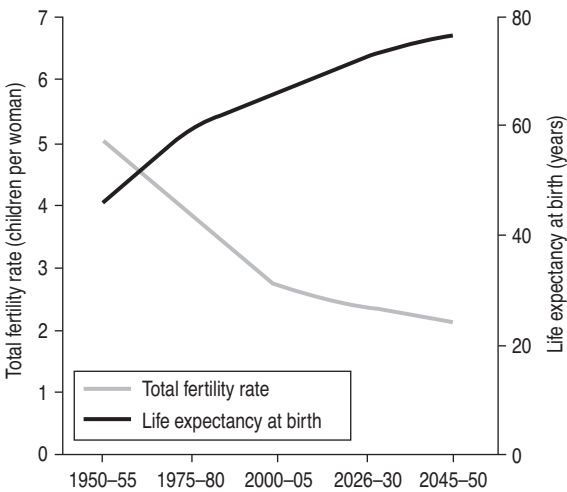
a	the average number of children a woman can expect to have.	developing countries
b	the average number of years a person born at a particular time is expected to live.	developed countries
c	the world's poorest countries	fertility rate
d	countries, which have high economic productivity, relatively high standards of living and relatively democratic systems of government	life expectancy

Why is the world's population ageing?

As life expectancy has increased and fertility rates have fallen both in developed and developing countries – although from a very different base – the world is experiencing an ageing of its population. According to the United Nations, Department of Economic and Social Affairs:

- In 2002, the number of people aged 60 years or older globally was estimated to be 629 million. By 2050 it is projected to reach 2 billion.
- One in 10 persons is now aged 60 years or over, by 2050 it is projected to be one in five, and by 2150 one in every three.
- Currently those aged 80+ make up 12% of the population aged 60 years or older. By 2050 it is projected that they will make up 19% of the population aged 60 years or older.
- Average life expectancy has increased 20 years since 1950. In 2002, average global life expectancy was 66 years.

Total fertility rate and life expectancy at birth, world, 1950–2050



[Source: US Bureau of the Census (2000) in UNFPA, *State of the World Population 2004*]

2a Calculate the expected decrease in the global fertility rate between 1950–55 and 2045–50.

2b Suggest reasons for the expected decrease in the fertility rate.

2c Calculate the expected increase in global life expectancy between 1950–55 and 2045–50.

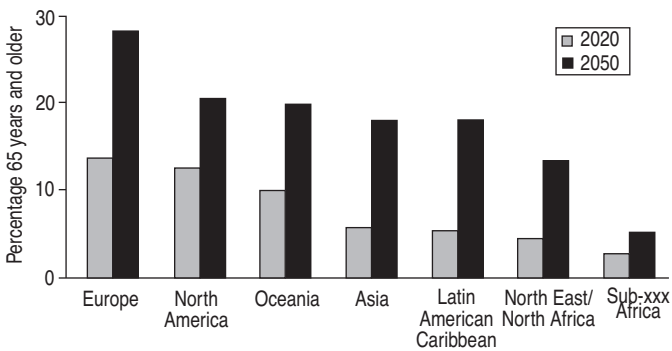
2d Suggest reasons for the expected increase in life expectancy.

Spatial distribution

Population aged over 60

	2002 (‘000)	2050 (‘000)
WORLD	628 874	1 963 767
More developed countries	235 523	395 106
Less developed countries	393 351	1 568 660
Least developed countries	34 419	175 713

Proportion of population over 65 by region, 2000 and 2050 (projected)



[Source: US Bureau of the Census (2000) in UNFPA, State of the World Population 2004]

- 3a Calculate the expected increase in the number of people globally over the age of 60 between 2002 and 2050.
- 3b Refer to the column graph and calculate the expected percentage increase in people over the age of 65 in each of the regions listed between 2000 and 2050.

Ageing facts – a summary

5 Mark the following statements true or false. Rewrite the incorrect statements correctly.

Fact or fiction?	T of F?	Correct statement
1 Falling life expectancy and increasing fertility rates both help to explain the ageing of the world's population		
2 The number of people worldwide over the age of 60 is expected to increase approximately threefold between 2002 and 2050.		
3 The ageing of the world's population will have economic and social consequences for both developing and developed nations.		
4 More developed countries will experience the greatest numerical increase in the number of people aged over 60 between 2002 and 2050.		
5 The region with the highest percentage of its population over the age of 65 is North America followed by Europe and Oceania.		
6 Less developed countries will experience the greatest fall in the potential support ratio between 2002 and 2050.		

Implications of an ageing population

The implications of an ageing population can be illustrated by the potential support ratio (PSR). That is, the number of persons aged 15 to 64 years per one older person aged 65 years or older.

According to the United Nations, Department of Economic and Social Affairs, the PSR has fallen and will continue to fall. Between 1950 and 2002, the PSR fell from 12 to 9 people. By 2050 it is projected to fall to 4.

Potential support ratio (PSR)

	2002	2050
WORLD	9	4
More developed countries	5	2
Less developed countries	12	5
Least developed countries	17	10

- 4a On a separate sheet explain what the potential support ratio is and its relevance.
- 4b Describe what is expected to happen to the potential support ratio globally between 2002 and 2050.
- 4c Discuss the implications of this change in the potential support ratio for developed and developing nations.
- 4d Describe the expected increase in the population aged 80 plus.
- 4e Summarise the factors that have led to the ageing of the world's population.

10 CHINA'S ONE CHILD POLICY

Introduction China's One Child Policy was introduced in 1979. Up until this time, the Chinese government had been encouraging its population to have children. However, by the 1970s a realisation had come about that if the country was going to develop and improve the economic circumstances of its people it would need to limit population growth.

Definitions

- quota** – a set amount
- population pyramid** – a bar graph showing via a series of horizontal bars the number or percentage of a population in a particular age group
- line graphs** – graphs that are used to show trends and change over time

China's population

China's population and projected population
1950–2050 ('000)

1950	562 580
1960	650 661
1970	820 403
1980	984 736
1990	1 148 364
2000	1 268 853
2010	1 347 563
2020	1 347 563
2030	1 461 528
2040	1 454 619
2050	1 424 162

Calculating percentage increase

Find the difference between two values and divide by the original number multiplied by 100

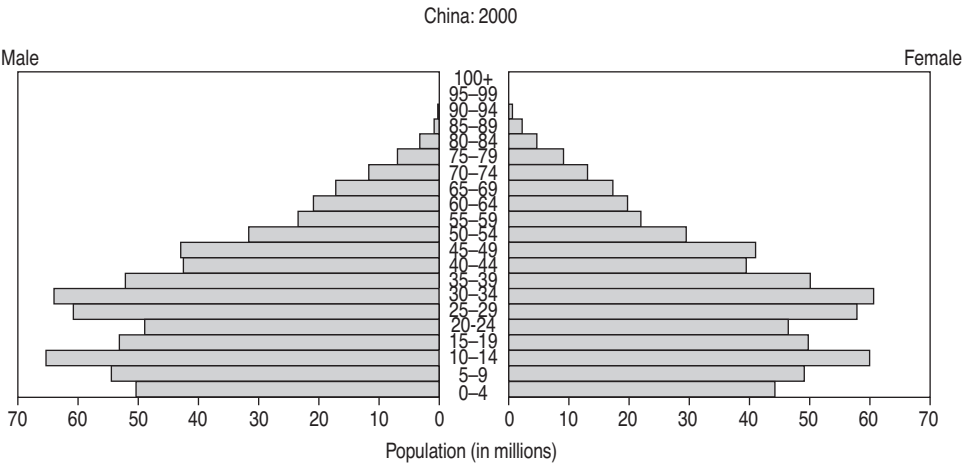
- 1** Refer to the statistics in the table at left to answer these questions:
- a** Calculate the percentage increase in China's population between 1950 and 1980.
- b** Calculate the expected percentage increase in China's population between 1980 and 2010.
- c** Comment on the difference in the percentage increase in China's population in these two periods.
- d** Construct a line graph on a separate sheet to show the data in the table.

[Source: US Census Bureau International Data Base]

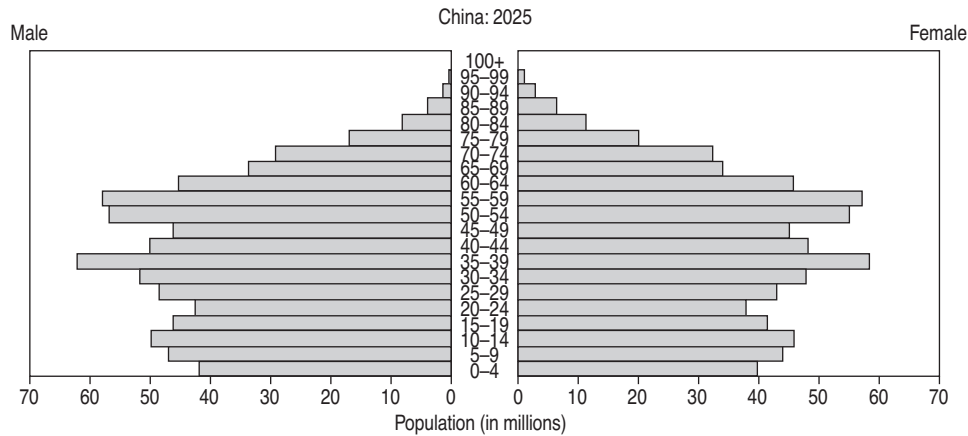
Interpreting a population pyramid

- In order to interpret a population pyramid there are a number of features of population pyramids that you should be aware of:
- Population pyramids display information about the age and sex structure of a total population.
 - A full pyramid represents 100 per cent of a population.
 - Population can either be shown as a percentage or a number – in the pyramids over it is a number.
 - Females are shown on one side of the graph and males are shown on the other.
 - Age categories are generally broken down into 5-year groupings, such as 0–4, and so on.
 - The scale of the graph is shown along the horizontal axis.

China population pyramid, 2000 [Source: US Census Bureau, International Data Base]



China population pyramid, 2025 (projected) [Source: US Census Bureau, International Data Base]



2a Describe the expected change in the age structure of China's population between 2000 and 2025.

2b Discuss the challenges that China is likely to face as a result of this changing age structure.

How does the One Child Policy work?

- Each workplace has a quota of the number of women who can get pregnant in a given year. Before having a child a woman must inform her work unit that she is planning on having a baby and will only be given permission if the yearly quota is not yet full.
- Incentives are used to encourage women to sign the One Child pledge such as an extra month's salary per year until the child is 14 or larger houses.
- Penalties are used if a woman breaks the One Child pledge such as a demotion at work or job loss, enforced abortions and sterilisations.
- If people have a second child without permission, it cannot be registered and therefore cannot legally attend school and will have difficulty in obtaining permission to marry in the future.
- While in cities most couples are restricted to one child, in some rural areas if the first child is a girl couples can seek permission to try a second time to have a son. The government accepts that in rural areas a son is needed to inherit the land and look after elderly parents.

Effects of the One Child Policy

- A significant drop in fertility. In the 1970s, China's fertility rate was 5.8 babies per woman, but is now at 1.7 babies per woman. Fertility has dropped to below the level to replace the population.
- A huge gender imbalance. Due to a cultural preference for sons, parents have used ultrasound tests to detect and selectively abort daughters. The latest census found 117 male newborns for every 100 females.
- Female infanticide. There have been many reports of baby girls being starved, abandoned or dumped.
- A generation of little emperors. Many people believe the One Child Policy has created a generation of pampered, over-indulged only children.
- The ageing of China's population.
- Human rights abuses. Many people feel that China's One Child Policy is an abuse of human reproductive rights.

3a Use the information in the two sections above to express your opinion on China's One Child Policy, its aims and operation.

3b Construct two mind maps:

- one titled: Positive Impacts of China's One Child Policy
- the other titled: Negative Impacts of China's One Child Policy.

3c Suggest some possible methods by which China's One Child Policy could be improved.

11 THE GLOBAL AIDS CRISIS

Definitions

AIDS – acquired immune deficiency syndrome, a disease in which the immune system is weakened and therefore less able to fight certain infections and diseases.

HIV – human immuno deficiency virus, the virus that causes AIDS

antiretroviral drugs – drugs that are designed to block the action of retro-viruses such as HIV. These drugs have successfully reduced the number of people dieing from HIV. They have stabilised their condition and enabled them to live at least another decade.

spatial distribution – how things are spread out over an area, in this case globally

Spatial distribution of AIDS

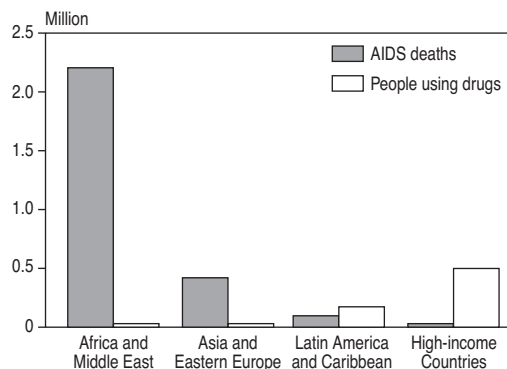
These are some of the significant facts about the AIDS crisis:

- At the end of 2002 the number of people living with HIV/AIDS was 42 million. [UNAIDS]
- At the end of 2001 more than 13 million children under the age of 15 in Africa, Asia and Latin America had lost a parent to AIDS. [UNICEF]
- More than 11 million of these children live in Africa. [UNICEF]
- The number of children who have lost 2 parents to AIDS is expected to increase from 3.8 million in 2001 to 6.9 million in 2010. [UNICEF]
- In 2002, average life expectancy in 16 African nations was at least 10 years lower than it would have been without AIDS. [UNAIDS]
- In Africa only 1% of the 4.4 million people in need had access to antiretroviral treatment. [WHO]

1a Use the points above to describe the spatial distribution of the global AIDS crisis.

1b Define the term 'pandemic' and use it in a sentence about AIDS.

AIDS deaths and number of people using antiretroviral drugs, by region, 2001



[Source HIV/AIDS Pandemic Spreads Further, Vital Signs 2003 pages 88–9, Worldwatch Institute]

2 Examine the graph above and complete the following:

a Estimate the number of people in Africa and the Middle East who died of AIDS in 2001.

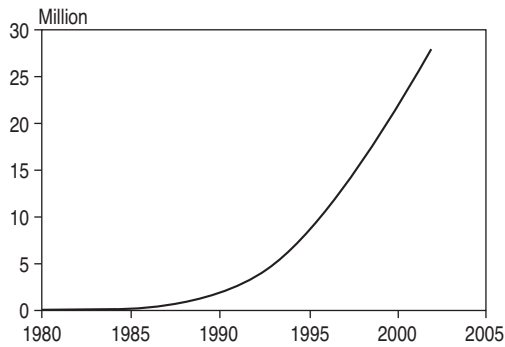
b Calculate the number who died of AIDS globally in 2001.

c Calculate the percentage of total global AIDS deaths in 2001 that the people who died of AIDS in Africa and the Middle East comprised.

d Describe the spatial distribution of the use of antiretroviral drugs globally. Suggest reasons for this spatial distribution.

Growing number of AIDS deaths

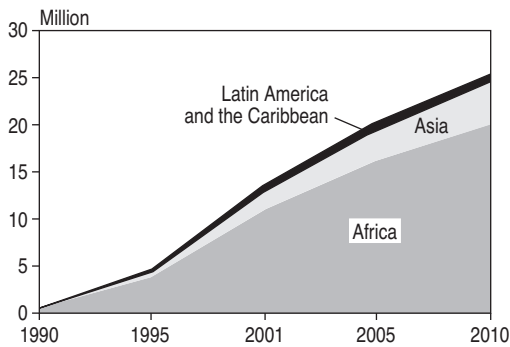
Estimates of cumulative AIDS deaths worldwide, 1980–2002



[Source: HIV/AIDS Pandemic Spreads Further, Vital Signs 2003, pages 68–9, Worldwatch Institute]

- 3 Use the line graph above to complete the following:
- a Calculate the increase in the number of people who died from AIDS between 1990 and 2000.
- b Calculate the percentage increase in the number of people who died of AIDS between 1990 and 2000.
- c Discuss why you think AIDS deaths increased so dramatically during this decade.

Orphans due to aids, by region, 1990–2001, with projections to 2010



[Source: Orphans Increase Due to AIDS Deaths. Vital Signs 2003 pp 110–11, Worldwatch Institute]

Calculating percentage increase

Find the difference between two values and divide by the original number multiplied by 100.

- 4 Refer to the graph showing Orphans Due to AIDS by Region to complete the following:
- a Estimate the number of children who were orphans due to AIDS in 2001.
- b Estimate the number of children expected to be orphans as a result of AIDS in 2010.
- c Calculate the expected percentage increase in the number of ophans due to AIDS between 2001 and 2010.
- d Estimate the number of orphans due to AIDS living in each of the following regions in 2001:
- Latin America and the Caribbean
 - Asia
 - Africa

The global AIDS crisis – a summary

- 5 Label the following statements true or false. Rewrite the incorrect ones correctly.

Fact or fiction?	T or F
AIDS is a virus that weakens the immune system.	
In 2002 there were 52 million people living with AIDS worldwide.	
The region with the highest incidence of AIDS is Asia.	
The AIDS virus is leading to lower average life expectancy in many African countries.	
Antiretroviral drugs for AIDS are more readily available in developing countries as they are being prioritised due to the large number of people with AIDS.	
Most of the orphans due to AIDS in 2001 were found in Africa.	

- 5b Draw up a table in your notes titled ‘The Global AIDS Crisis’. Summarise the causes and impacts of the crisis and suggest solutions.

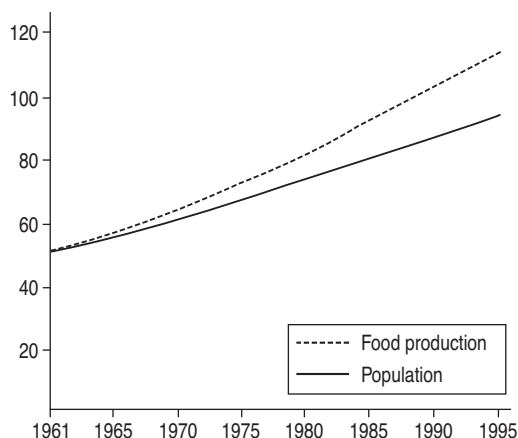
Causes	Impacts	Solutions

12 WORLD FOOD SUPPLIES

Introduction World food production has continued to expand during the last century to provide for a growing population. Despite this, the FAO (United Nations Food and Agricultural Organisation) reported in 2003 that an estimated 850 million people still experience regular significant to severe food shortages.

Food supplies and population growth

Population and food production (rate of change index based on 1961 levels)



[Source: FAO]

- 1a** Describe the changes in global food production and population since 1961.
- 1b** Explain how it can be possible for so many to still experience food shortages and starvation, despite the trend shown in the graph. You could answer with general points or concepts, and may be able to illustrate them with examples from current affairs or recent knowledge of world issues.
- 1c** List the difficulties that the FAO might encounter in gathering and measuring data on global food production.

What the world eats

2 Examine the table at top right then answer these questions:

- a** Describe briefly, and in general, what this table shows.
- b** The data is supplied by the FAO. What do these letters stand for? What larger agency is this organisation a part of?
- c** Prepare a table or list, ranking the foods for each group in descending order.
- d** Using examples compare and contrast the diet of people in the developed and developing worlds. Include both total and composition of food.

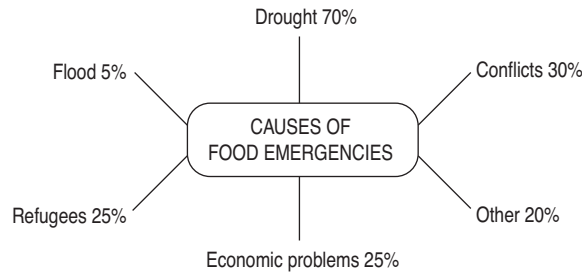
World food intake – food types 1999–2001

Food type	Developed world %	Developing world %
Cereals	30.9	53.6
Roots and tubers	4.1	5.6
Sugar	12.8	7.5
Pulses – dry (peas, beans lentils, etc)	0.9	2.3
Vegetable oils	13.9	10.7
Meat	10.2	6.6
Milk /dairy	8.7	2.8
Other	18.6	10.9
Total food supply in calories per capita per day	3261	2675

[Source – FAO, 2003]

- e** Suggest reasons for similarities and differences in the diets of peoples in the two sections of the table.
- f** Which group do you consider has the best diet? Give reasons for your answer.
- g** What issues of food storage are evident in the table? (consider perishability)
- h** Which group would source its food from a larger geographical area? Give reasons for your answer.
- i** Different foods require different types and volumes of resources (land, capital, labour, enterprise) to produce. Outline the difference in resources that would need to be used by each group for food production.
- j** From the table, the average person in the Developing world consumes 82% of the calories consumed by someone in the Developed World. Do you consider that the world's food problems have therefore largely been solved?

Problems producing food



The main causes of food emergencies in developing countries in 2002 according to the FAO are shown in the diagram above. (Note that the total exceeds 100% as multiple causes are cited for most emergencies)

3a Explain how each of the problems in the diagram above would affect food production or lead to a food emergency.

3b What incidents or conditions can you think of that might be placed in the 'Other' category.

The FAO reports

Despite international efforts to reduce poverty, hunger is on the rise again after falling steadily during the first half of the 1990s, according to the annual FAO report, 'State of Food Insecurity in the World'.

Almost 850 million people go to bed hungry every night, the vast majority in Africa and Asia, while the number of hungry people in the developing world is climbing at a rate of almost 5 million a year, according to the UN.

At the UN World Food Summit in 1996, governments set a goal of cutting in half the number of hungry people by 2015, using 1990–2 as a base-line. The FAO said it was time nations examined why hundreds of millions of people went hungry in a world that produced more than

enough food for everyone. 'Bluntly stated, the problem is not so much a lack of food as a lack of political will,' the report says.

Except when wars or natural disasters put a spotlight on developing countries, 'little is said and less is done' to end the plight of the 798 million people who suffer from chronic hunger, the report says.

Another 11 million people in the developed countries and 34 million in countries in transition are hungry.

According to the FAO report, during the first half of the 1990s, the number of chronically hungry people decreased by 37 million. But since 1995–7, the number has increased by 18 million. A closer analysis of the figures revealed 'an even more alarming trend' – that the number of

hungry people in the developing world increased by 4.5 million a year between 1995 and 2001.

About 19 countries reduced the number of hungry people since 1990–2 by a total of over 80 million. Twenty-two countries, including Bangladesh, Haiti and Mozambique 'turned the tide' against hunger, in the second half of the decade.

At the other end of the scale are 26 countries where hunger increased by almost 60 million, including Afghanistan, Congo, Burundi, North Korea, Somalia, Tanzania, Guatemala, Liberia and Sierra Leone.

[Adapted from the FAO Report, 'State of Food Insecurity in the World', 2003]

4a Who produced the report on world food insecurity?

4b How often is it produced?

4c What was the goal set by the World Food Summit in 1996?

4d Explain the statement: 'the problem is not so much a lack of food as a lack of political will.' (Note: There was 15% more food per capita available globally in 2000 than in 1980!)

4e Explain, using examples, how wars and natural disasters lead to hunger. Give examples of recent events and conflicts in your response.

4f Construct a Sector (pie) Graph to show the distribution of people in hunger in areas of differing development.

4g On both a global scale and at a national level, it is the distribution of food between people that is the real issue. Use evidence from the article to support this statement.

4h Is the world making progress toward the war on hunger? Use evidence from the article to support your response.

4i Use an atlas to locate the countries cited in the article. What spatial patterns are evident?

4j If you were a national or UN leader, what would you do help solve the world's hunger problem. Your suggestions may differ according to the country that you come from. Consider all the causes of hunger, the volume and distribution of current food supplies and the size, growth and distribution of the world's people.

13

URBANISATION

Note to teacher

Students will need to use an atlas and a blank outline map of the world to complete this worksheet.

Introduction Throughout the 20th century, there was a rapid increase in the proportion of people living in urban areas. As a result, by the end of the 20th century – for the first time in history – there were more people living in towns and cities than in rural areas. The rapid rate of urbanisation has led to many problems as cities struggle to cope with dramatically increased numbers.

Definitions

developing world – the world’s poorest countries

urbanisation – the process by which an increasing proportion of a country’s or region’s population are living in towns and cities

rural–urban migration – the movement of people from rural areas to the cities due to push and pull factors

natural increase – the difference between the birth rate and the death rate in a population

megacities – large urban agglomerations with at least 10 million inhabitants

spatial distribution – where things are

The process of urbanisation

Urban population as a % of total population				Urban population annual growth rate (%)	
	1970	1995	2015	1970–1995	1995–2015
Least developed	12.7	22.9	34.9	5.8	4.6
All developing	24.7	37.4	49.3	3.8	2.9
Industrialised	67.1	73.7	78.7	1.1	0.6

[Source: UNDP Human Development Report, 2000]

1 Use the information in the table to fill in the blanks below.

Between 1970 and 2015 the urban population as a percentage of the total population in the world's developing countries is expected to increase by _____, while the level of urbanisation in industrialised countries is expected to increase by _____. The reason the increase in the proportion of people living in urban areas is much higher in _____ than in _____ is because the level of urbanisation was already high in _____ in 1970. The least developed countries are also expected to experience a significant increase in urbanisation. However this increase will not be quite as _____ as for developing countries as a whole. The rate of growth of the urban population is expected to be _____ in all regions of the world in 1995–2015 than it was in _____.

Causes of urbanisation

The two factors that have led to urbanisation are:

- rural–urban migration
- natural increase

Push factors	Pull factors
Famine	Employment

2a Construct a table like the one at left. Make a list of the reasons why people are leaving rural areas (push factors) and why they are moving to urban areas (pull factors). An example has been done for you.

2b Suggest reasons as to why many developing countries may have experienced an increase in natural increase.

The rise of megacities

A feature of the urbanisation process has been the rise of megacities. Increasingly megacities are concentrated in the developing world.

Population of cities with 10 million or more inhabitants

Rank	1950 City/population (million)	1975 City/population (million)	2001 City/population (million)	2015 City/population (million)
1	New York 12.3	Tokyo 19.8	Tokyo 26.5	Tokyo 27.2
2		New York 15.9	Sao Paulo 18.3	Dhaka 22.8
3		Shanghai 11.4	Mexico City 18.3	Mumbai 22.6
4		Sao Paulo 10.3	New York 16.8	Sao Paulo 21.2
5			Mumbai 16.5	Delhi 20.9
6			Los Angeles 13.3	Mexico City 20.4
7			Calcutta 13.3	New York 17.9
8			Dhaka 13.2	Jakarta 17.3
9			Delhi 13.0	Calcutta 16.7
10			Shanghai 12.8	Karachi 16.2
11			Buenos Aires 12.1	Lagos 16.0
12			Jakarta 11.4	Los Angeles 14.5
13			Osaka 11.0	Shanghai 13.6
14			Beijing 10.8	Buenos Aires 13.2
15			Rio de Janeiro 10.8	Metro Manila 12.6
16			Karachi 10.4	Beijing 11.7
17			Metro Manila 10.1	Rio de Janeiro 11.5
18				Cairo 11.5
19				Istanbul 11.4
20				Osaka 11.0
21				Tianjin 10.3

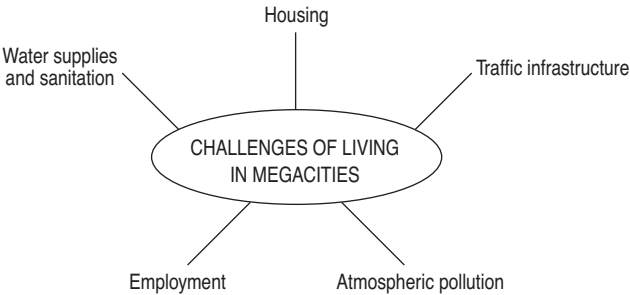
[Source: United Nations Population Division, World Urbanisation Prospects: The 2001 Revision]

Calculating percentage increase:

Find the difference between two values and divide by the original number multiplied by 100.

- 3 Refer to the table above to complete the remainder of questions and activities on this page.
- a Locate the world's mega cities in 2001 on a blank outline map of the world. Use a different colour dot for those in the developed world and those in the developing world.
- b Describe the spatial distribution of mega cities in 2001 and the expected spatial distribution of mega cities in 2015.
- c Identify which megacities were located in the developing world in 2001.
- d Calculate the proportion of the megacities globally, which were located in the developing world in 2001.
- e Calculate the percentage increase in the population of Sao Paulo and Shanghai between 1975 and 2001.
- f Calculate the average annual percentage population increase of Sao Paulo and Shanghai between 1975 and 2001.

- g Design a concept map titled 'Challenges of Living in Megacities'. Use the diagram below as a starting point.
- 4 You have been employed by the United Nations Development Program to conduct a study and write a report



on how life could be improved for slum dwellers in the mega cities of the developing world. Your report should be up to two pages in length and address the challenges which you have identified in your concept map above.

14 FORCED MOVEMENT OF PEOPLE

Introduction In 2002, there were approximately 20 million people worldwide, who had been identified by the UNHCR as persons of concern. They included refugees, asylum-seekers and displaced persons.

Definitions

1 Match these terms with their definitions in the table below: refugee, asylum seeker, convention, UNHCR, persons of concern, internally displaced person (IDP)

	1 a person seeking refuge
	2 people who have lost their homes but remain inside their country
	3 stateless and war-affected populations
	4 someone with a well founded fear of persecution on the basis of their race, religion, nationality, a member of a particular social group or political opinion who is outside their country of nationality and unable or unwilling to return
	5 United Nations High Commissioner for Refugees
	6 an agreement

Refugees and persons of concern

Estimated number of refugees and total persons of concern to UNHCR worldwide

Year	Refugee estimate	Total persons of concern
1980	8 439 000	
1985	11 844 000	
1990	17 370 000	
1995	14 855 000	26 103 000
2000	12 062 000	21 814 000

[Source: UNHCR 2001 Global Population Statistics]

2a Construct a bar graph next to the table above to show the number of refugees in each of the years shown in the table. Add a heading.

2b Suggest reasons why there was such a high number of refugees globally in 1990.

2c Explain why the UNHCR has widened their mandate to include persons of concern and not just refugees.

Persons of concern to UNHCR (1 January 2002, by category)

Region	Refugees	Asylum-seekers	Returned refugees	IDPs and others of concern	Total 1 Jan 2002
Asia	5 770 300	33 100	49 200	2 968 000	8 820 700
Africa	3 305 100	107 200	266 800	494 500	4 173 500
Europe	2 227 900	335 400	146 500	2 145 600	4 855 400
Northern America	645 100	441 700	–	–	1 086 800
Latin America & Caribbean	37 400	7 900	200	720 000	765 400
Oceania	65 400	15 600	–	300	81 300
Total	12 051 100	940 800	462 700	6 328 400	19 783 100

[Source: UNHCR 2001 Global Population Statistics]

2d Construct a pie graph next to the table above to show each region's share of the world's refugees. Add a heading.

Challenges for refugees

- 3 Complete the table below. Identify the difficulties refugees may experience in leaving their own country and the challenges they may face in their country of asylum.

Difficulties leaving	Challenges in new country

Impact of refugees on their country of asylum

- 4 Complete this table by listing the positive impacts refugees can have on their country of asylum and the difficulties refugees could create, then make suggestions for how these difficulties could be overcome.

Positive impacts	Difficulties	How difficulties can be overcome

Facts about refugees

- 5 Label the following statements true or false.

Fact or fiction?	T or F?
1 A person, who leaves a country voluntarily to seek a better life in a new country would be considered a refugee.	
2 In January 2002, 18% of the world's refugees were from Africa.	
3 An internally displaced person is someone who has lost their home but remains inside their country.	
4 A person who has fled their country of origin because they were tortured for speaking out against their government would be considered a refugee.	
5 A group of people, who have fled political persecution, but remain within their country are known as asylum seekers.	
6 The region with the highest number of refugees in 2002 was Asia, closely followed by Europe.	
7 The number of refugees worldwide has continuously increased since 1980.	

Protecting refugees – the Geneva Convention

The 1951 Geneva Convention relating to the status of refugees is the key legal document in defining a refugee. It outlines the rights and responsibilities of both the refugees and the states that are parties to the convention. It was originally focussed on Europe, then strengthened and expanded worldwide by the 1967 protocol. By 2004 there were 142 states that were parties to the convention.

- 6 Use the internet to conduct research into the 1951 Geneva Convention on refugees. Outline the responsibilities of nations who are signatories to the convention.

15 MEASURING DEVELOPMENT

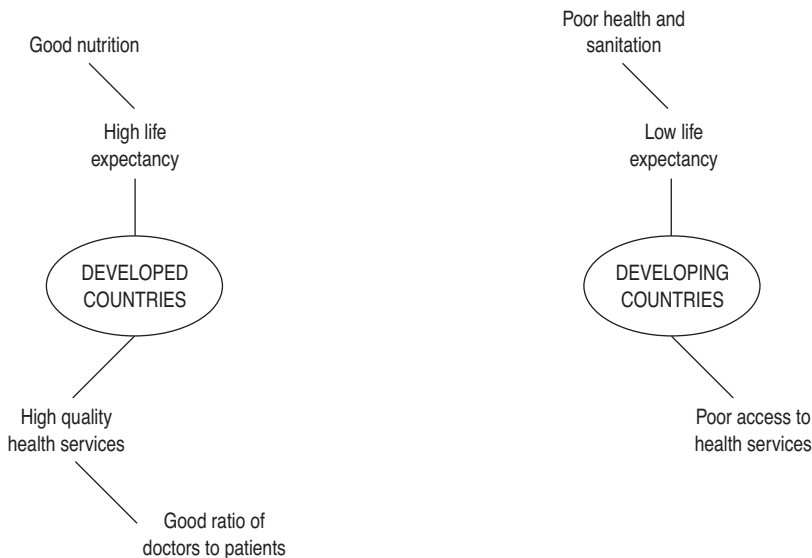
Introduction Human development varies enormously on a global scale. There are huge disparities in the quality of life of individuals in areas such as health, education and income. In order to gain an understanding of the spatial patterns of human development, geographers use development indicators that examine various aspects of the economic, social, environmental or political characteristics of a nation.

Note to teacher
Students will need to use an atlas and a blank outline map of the world to complete this worksheet.

Definitions
developing countries – the world’s poorest countries
developed countries – countries that have high economic productivity, relatively high standards of living and relatively democratic systems of government

Characteristics of development

1a Develop a concept map of the characteristics of **developed** and **developing** countries on a separate sheet using the diagrams below as a starting point.



1b Write your own definition of what you believe a developed nation is. In writing your definition you should consider the economic, social, political and environmental factors.

Indicators of development

2a The table below and continued over defines some examples of measures of development. Match the following terms with the correct definition in the table.

adult literacy rates, GDP, GDP per capita, Gini index, Human Development Index, under five mortality, life expectancy, maternal mortality ratio, purchasing power parity

Measure of development	Term	Rank
1 The percentage of adults who can read and write.		
2 The average number of years a person born at a particular time is expected to live.		
3 The total value of goods and services produced in a country in a year.		
4 An index that measures the extent to which the distribution of income (or consumption) among individuals or households in a country deviates from perfectly equal distribution. A value of 0 represents perfect equality; a value of 100 perfect inequality.		

5	The total value of goods and services produced in a country in a year divided by the total population of a country.		
6	A rate of exchange that accounts for price differences across countries, allowing international comparisons of real output and incomes. PPP US\$1 has the same purchasing power in the domestic economy as \$1 in the United States.		
7	An index that combines scores on income per head, literacy rates, school enrolment and life expectancy to produce a figure between zero and one, where one is the highest score.		
8	The probability of dying between birth and exactly five years of age, expressed per 1000 live births.		
9	The annual number of deaths of women from pregnancy related causes per 1000 live births.		

2b Rank the above indicators from 1 to 9, with 1 being the indicator that you think gives the best measure of how developed a country is and 9 being the least useful method of how developed a country is.

2c Write a paragraph justifying your rank order.

Development in Asia

	Life expectancy (years)	Under 5 mortality male/female	GDP per capita (PPPUS\$)	Access to safe water
Eastern Asia				
China	71	39/47	4 580	75
Democratic People's Republic of Korea	63	61/55		100
Hong Kong SAR	80	5/5	26 910	100
Japan	82	5/4	26 940	100
Mongolia	64	88/83	1 710	60
Republic of Korea	76	8/6	16 950	92
South-East Asia				
Cambodia	57	115/99	2 060	30
Indonesia	67	59/46	3 320	78
Lao People's Democratic Republic	55	144/137	1 720	37
Malaysia	73	15/11	9 120	100
Myanmar (Burma)	57	137/118	1 027	72
Philippines	70	40/30	4 170	86
Singapore	78	4/4	24 040	100
Thailand	69	31/19	7 010	84
Vietnam	69	52/37	2 300	77
South Central Asia				
Afghanistan	43	278/283		13
Bangladesh	61	85/90	1 700	97
Bhutan	63	82/78	1 969	62
India	64	78/90	2 670	84
Iran	70	39/39	6 690	92
Nepal	60	91/106	1 370	88
Pakistan	61	121/135	1 940	90
Sri Lanka	73	30/16	3 570	77

[Source: United Nations Human Development Report, 2004 and The State of the World Population, 2004]

3a Using each indicator, classify the above countries as either having low, medium or high levels of development. (You will need to work out a cut off point for each category.)

3b Compare your classifications of the countries using each of the 4 indicators. Is there a correlation in how each of the countries were classified using each of the different indicators? Write an analysis.

3c Select an indicator and construct a choropleth map of show its variation throughout the Asian region.

16

INTERNATIONAL AID

Introduction International Aid comes from a variety of government and non-government sources and exists in a range of forms from financial assistance at the national level to food and clothing in a disaster at the local level. However, it is a controversial aspect of human development. Criticisms include concerns over the amount of aid being given by nations, approaches to providing aid, lack of accountability, the donor benefiting more than the recipient and the imposition of a Western model of development on developing nations.

Definitions

GNI – gross national income. The sum of the value added by all resident producers in an economy, plus receipts of primary income from foreign sources.

OECD – Organisation of Economic Cooperation and Development

Types of aid

1 Match each term in the following list with the correct definition in the table.

bilateral aid, emergency aid, multilateral aid, non-government organisation, official development assistance, project aid, tied aid, untied aid

1	Aid given directly from government to government.	
2	Aid from governments indirectly to recipient governments through multilateral institutions such as the World Bank.	
3	Aid in the form of food, clothing and medicines following disasters.	
4	Projects ranging from the construction of large infrastructure such as dams and bridges to small scale technical assistance being provided.	
5	A voluntary not for profit organisation that accepts donations from the public and provides aid to developing countries through specific projects.	
6	Grants, loans, goods and services given by the government of developed nations to assist the development of developing countries.	
7	Where a country receiving aid in the form of money is free to spend it in a way that they choose.	
8	When the donor country sets conditions on how the recipient country must spend the aid.	

Why do countries give aid?

2 Why do countries give aid? Draw up a table like the one below in your notes. List as many economic, social and political reasons for giving aid as you can think of. Consider both the benefits to the donor and recipient nation.

Benefits to donor nation	Benefits to recipient nation

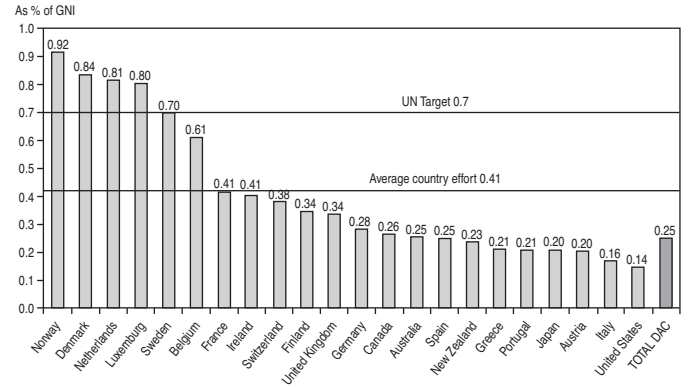
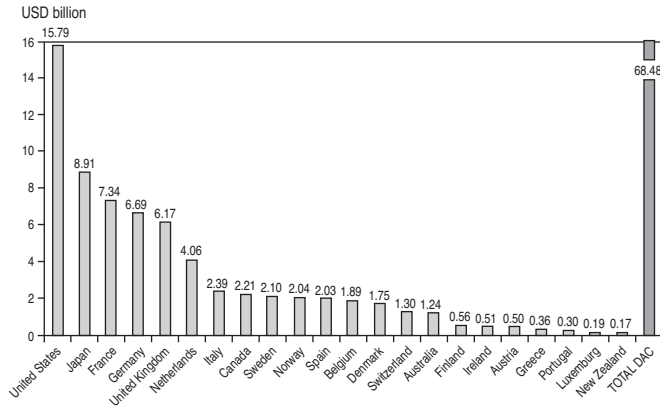
Official development assistance

The United Nations has set a target for overseas aid called ODA (official development assistance). The ODA is set at 0.7% of a developed country's GNI. However, only a small number of nations actually donate this percentage, as can be seen from the graphs on the next page.

- 3a Use the graphs to list the top 3 OECD donors of official development assistance (ODA) in dollar terms.
- 3b Where does the official development assistance of each of these 3 countries rank as a percentage of GNI?
- 3c How many OECD countries actually gave the United Nations recommended 0.7% of GNI in 2003?
- 3d List the names of these countries.

Net Official Development Assistance, OECD countries, 2003

[Source: OECD, 16 April 2004]



Australia's aid

Top 6 countries for Australian government aid 2004–05

Country	\$AUD (millions)
Papua New Guinea	435.6
Solomon Islands	201.6
Indonesia	160.8
Vietnam	73.7
Philippines	62.2
China	49.3

[Source: AUSAID, 2004]

- 4a** Label the top 6 countries for Australian Government Aid 2004/05 on a map of the world.
- 4b** Describe the spatial distribution of the top 6 recipients of Australia's aid.
- 4c** Explain the possible reasons these countries are the top recipients of Australia's aid.

Non-government organisations

- 5a** List the names of non-government organisations in Australia that give aid overseas.
- 5b** Use the internet to conduct research into a non-government organisation of your choice. Outline its aims, approach and activities.
- 5c** Evaluate the effectiveness of the non-government organisation in achieving its aims.

Appropriateness of aid

6 Read the quote in the next column from Oxfam Community Aid Abroad's Submission to the Human Rights Sub-Committee of the Joint Standing Committee on Foreign Affairs Defence and Trade 'The link between aid and human rights'. Answer the following questions.

- a** Discuss why the provision of aid is so problematic.
- b** From the quote below list the things that need to be avoided when providing aid.
- c** Suggest ways that aid can be provided in an appropriate, effective and culturally sensitive way.

Charity Adage

'Give a man a fish and you feed him for a day, teach a man to fish and you feed him for a lifetime.'

Human Rights View From The Mekong River

'A man' is just as likely to be 'a woman', and she already knows how to fish.

She would like her river left alone by illegal logging companies or fish poachers using dynamite and working in collusion with corrupt local government officials.

She would prefer that her government did not build huge dams with the help of the Asian Development Bank, dams that have reduced fish stocks dramatically and damaged her livelihood.

She would prefer that police did not use violence to enforce mass evictions to enable the dams to be built.

She doesn't want charity.

She would like her basic rights respected.

[The sentiments of Thin, a woman leader from Check village, Takeo Province, partner in the Community Aid Abroad-Oxfam Australia program in Cambodia.]

17

DIVERSITY IN ASIA

Note to teacher

Students will need to use an atlas and a blank outline map of the world to complete this worksheet.

Introduction As the following activities illustrate, the region we call Asia is incredibly diverse. It is made up of a range of peoples living in different environments, speaking different languages and living in very different economic circumstances.

Definitions

GDP per capita – the total value of goods and services produced in a country in a year divided by the total population of a country.

(PPP US\$) – a rate of exchange that accounts for price differences across countries, allowing international comparisons of real output and incomes. PPP US\$1 has the same purchasing power in the domestic economy as \$1 in the United States.

Location and climate

- 1a** Using an atlas mark each capital city on a blank outline map of Asia.
- 1b** In the second column in the table below write the name of the country to which each capital city belongs.
- 1c** From the table below it is evident that climate varies across Asia according to latitude. Using evidence from the table explain how and why climate varies according to latitude.

Capital city	Country	Elevation (metres)	Latitude	Longitude	Average temperature °C
Beijing		54	39°56'N	116°17'E	12
Tokyo		7	35°33'N	139°47'E	15
Kathandu		1336	27°42'N	085°22'E	18
Taipei		6	25°04'N	121°32'E	22
Hanoi		5	21°01'N	105°48'E	24
Mubai		14	19°07'N	07°51'E	27
Vientiane		170	17°57'N	102°34'E	26
Phno Penh		11	11°33'N	104°51'E	27
Jakarta		29	06°15'S	106°54'E	27
Dili		6	08°34'S	125°34'E	25

Who am I?

- 2** How well do you know Australia's nearest neighbours? Identify the country from the description below:
- a** I am a country made up of 13 600 islands. I have a population of 220 million people from over 300 ethnic groups. I was colonised by the Dutch but gained independence in 1949. I am now a republic with President Susilio Bambang Yudohoyono as my Head of State. _____
- b** I am the most populous nation in the world with a population of 1.3 billion. My people speak Cantonese and Mandarin. In recent years I have experienced dramatic rates of economic growth. _____
- c** I am South-East Asia's newest nation. I am a tiny country with a land area of 15 007 square kilometres. I have a population of 997 853. My main industries are agricultural including coffee, rice and maize. My people are not wealthy. My GDP per capita is US\$500. I am hoping to improve my economy in the future through the development of oil and natural gas resources. _____
- d** My land area is comprised of 7000 islands. I have a population of 84 million people who are predominantly descendants of Malay, Chinese and Muslim minorities. The majority of my people are Catholic due to the influence of Spanish colonialism. I am a republic. My Head of State is Gloria Macapagal-Arroyo. _____

Population

3 Use lines to match each country with the correct population listed in the second column.

China	4.3 million
Japan	1.08 million
Singapore	19.2 million
Vietnam	127.8 million
India	82.5 million
Sri Lanka	1.3 billion

Language

4 Use lines to match each language with the country in which it is spoken.

Mandarin	Sri Lanka
Tagalog	East Timor
Sinhalese	China
Tetun	Laos
Lao	Burma (Myanmar)
Kachen	Philippines

Colonisation in Asia

5 Many of the countries of Asia were colonised by European powers. These European powers had a huge impact on all aspects of life in the countries that they colonised.

a Match each of the following nations with the country that colonised it.

Indonesia	Portugal
Philippines	France
Viet Nam	Spain
Malaysia	Netherlands
East Timor	Britain

b Discuss how the colonisers may have influenced the nations that they colonised.

Economy

6 Refer to the following table:

- a Explain how GDP per capita PPP \$US can be used as an indicator of development.
- b In the last column rank the countries of Asia from the most developed to the least developed according to GDP (PPP \$US).
- c Construct a choropleth map to show the variations in GDP (PPP \$US).

The economies of Asia, 2003

	GDP per capita (PPP US\$)	Rank
Eastern Asia		
China	4 580	
Democratic People's Republic of Korea*	1000	
Hong Kong SAR	26 910	
Japan	26 940	
Mongolia	1 710	
Republic of Korea	16 950	
South-East Asia		
Cambodia	2 060	
Indonesia	3 320	
Lao People's Democratic Republic	1 720	
Malaysia	9 120	
Myanmar	1 027	
Philippines	4 170	
Singapore	24 040	
Thailand	7 010	
Vietnam	2 300	
South Central Asia		
Afghanistan*	700	
Bangladesh	1 700	
Bhutan	1 969	
India	2,670	
Iran	6 690	
Nepal	1 370	
Pakistan	1 940	
Sri Lanka	3 570	

[Source: United Nations Human Development Report, 2004]

[* North Korea and Afghanistan are 2003 estimates from the CIA Factbook]

Constructing a choropleth map

1

Decide on an appropriate 4- or 5-hierarchical classification (numerical range).

2

Construct a key.

3

Select a colour for each numerical range to be shown on the map.

4

Shade the key.

5

Shade the relevant areas on the map the appropriate colour.

18 CULTURAL INTEGRATION

Note to teacher

Students will need to use an atlas and a blank outline map of the world to complete this worksheet.

Introduction Cultures have always influenced one another. However in recent decades with the greater movements of people, advances in technology and growth in the importance of trade in the global economy as well as the rise of global media, the process of cultural integration has accelerated.

The spread of global culture

1 Develop a mind map of the various ways global culture has been diffused, adopted and adapted. Use the diagram at right as a starting point.

Food

The spread of global culture in Australia is particularly evident from the incredible variety of food available.

2a Match the following foods eaten in Australia with their country of origin.

	Food	Country of origin
A	dolmathes	China
B	gado gado	Greece
C	Fried rice	India
D	kaftah	Indonesia
E	kebab	Italy
F	laksa	Japan
G	naan bread	Lebanon
H	pasta	Malaysia
I	Sushi	Thailand
J	Tom Yum Soup	Turkey

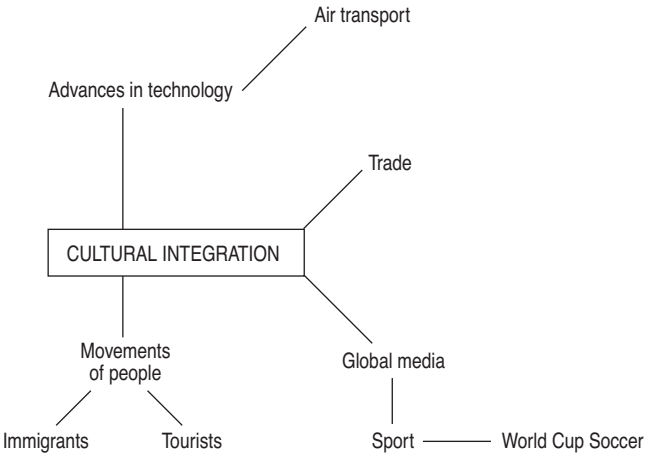
2b Add another five examples of foods and their country of origin to the table.

2c Consider how these foods have come to Australia.

2d List 2 examples of how Australians have adapted the foods of other countries.

Definitions

- cultural integration** – the diffusion, adoption and adaptation of culture globally
- diffusion** – spreading out over a large area
- adoption** – taking up and embracing
- adaptation** – changing to suit



Language

Culture has spread globally through language. French is just one example.

3a Mark the French-speaking countries in the list below on a map of the world.

Benin, Burkino-Faso, Burundi, Cameroon, Canada, Central African Republic, Chad, Comoros, Congo, Cote d'Ivoire, Democratic Republic of Congo, Djibouti, France, Gabon, Guinea, Haiti, Luxembourg, Madagascar, Mali, Monaco, New Caledonia, Niger, Rwanda, Senegal, Seychelles, Togo, Vanuatu, Vietnam

3b Describe the spatial distribution of French-speaking countries.

3c Describe how the French language may have impacted on the culture of these countries.

Movement of people

One of the most obvious ways in which cultural influences are diffused, adopted and adapted is through the movement of people both past and present. The languages other than English that are spoken at home indicate the cultures that have impacted on Australian society.

- 4 The table below lists the 6 most common languages other than English spoken at home by Australians in 2001. Draw up a table like the one below in your notes.
- a In the second column, brainstorm specific examples of ways that the culture represented by each of these languages has influenced and impacted on Australian society. (Consider factors such as food, fashion, sport, education, etc.)
- b In the third column list examples of ways that Australians have adapted aspects of these cultures.

Culture	Impact of culture	Adaptation of culture by Australian society
Italian		
Greek		
Cantonese		
Arabic (incl. Lebanese)		
Vietnamese		
Mandarin		

Technological change

- 5 Draw up a table like the one below in your notes.
- a In the second column place the correct invention with the corresponding date: aeroplane, internet, personal computer, telephone, television.
- b In the third column describe how each technological invention has contributed to the spread of global culture.

Year	Name of invention	Contribution to cultural integration
1875		
1903		
1930s		
1974		
1983		

Trade

The breakdown of trade barriers has seen an increase in global trade. Trade now comprises an increased proportion of global economic activity.

- 6a Examine the table opposite and calculate the increased trade in goods between 1990 and 2002 in each category.
- 6b List examples of how increased trade may have led to cultural integration.

Trade in goods

	% of GDP 1990	% of GDP 2002	Increase
WORLD	32.5	40.3	
Low income	26.9	37.3	
Middle income	35.2	54.9	
Lower middle income	30.6	49.2	
Upper middle income	45.0	66.2	
Low & middle income	33.4	51.8	
East Asia & Pacific	47.0	63.4	
Europe & Central Asia	28.8	64.3	
Latin America & Carib.	23.1	41.2	
Middle East & N Africa	46.6	50.5	
South Asia	16.5	24.2	
Sub-Saharan Africa	40.8	55.3	
High Income	32.3	37.6	
Europe EMU	44.9	56.3	

Note to teacher

Students will need to use an atlas and a blank outline map of the world to complete this worksheet.

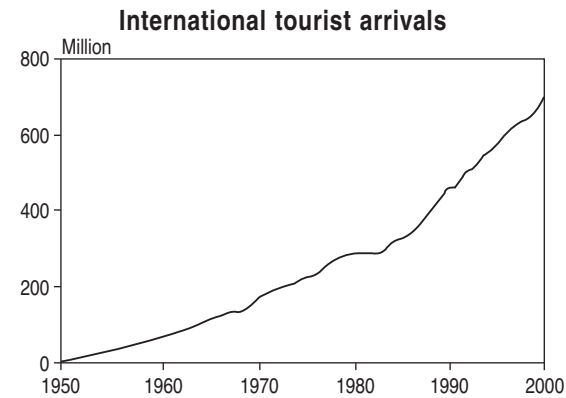
Introduction Tourism is defined by the World Tourism Organisation (WTO), as the activities of people who travel outside their usual environment for no more than a year for leisure, business or other purposes. The industry has grown dramatically since 1950 and it is now one of the world’s largest industries and biggest employers.

Definitions

- international tourism receipts** – amount of money received
- spatial distribution** – how things are spread out over an area, in this case globally

Tourists on the move

- 1 Refer to the graph at right to complete the following:
- a Calculate the increase in international tourist arrivals from 1950 to 2000.
- b Calculate the percentage increase in international tourist arrivals from 1950 to 2000.
- c List possible reasons as to why the tourism industry has grown so dramatically since 1950.
- d List as many reasons you know of why people travel. Rank your completed list in the order of importance you give to each reason. Would other people have the same ranking? Explain.



World’s top 10 destinations by international tourism receipts

Rank		2002 (US\$ billion)
1	USA	66.5
2	Spain	33.6
3	France	32.3
4	Italy	26.9
5	China	20.4
6	Germany	19.2
7	UK	17.8
8	Austria	11.2
9	Hong Kong	10.1
10	Greece	9.7
	WORLD	

[Source: World Tourism Organisation Tourism, Highlights Edition 2003]

Tourist destinations

World’s top 10 destinations by arrivals

Rank	Destination	2002 (million)
1	France	77.0
2	Spain	51.7
3	USA	41.9
4	Italy	39.8
5	China	36.8
6	UK	24.2
7	Canada	20.1
8	Mexico	19.7
9	Austria	18.6
10	Germany	18.0
	WORLD	703.0

[Source: World Tourism Organisation, Tourism Highlights Edition 2003]

- 2 Refer to the two preceding tables to complete the following:
- a Locate the top 10 tourism destinations by arrivals on a blank outline map of the world.
- b Describe the spatial distribution of the world’s top tourism destinations by arrivals.

- c** Suggest possible reasons for the spatial distribution of the World's Top Tourism Destinations by arrivals.
- d** Discuss possible changes that are likely to occur to this list of top 10 tourism destinations by arrivals in the next decade and explain why you think these changes may occur.
- e** Calculate the percentage share of total world tourism arrivals of each of the following countries in 2002: France, Spain, USA.
- f** France is the world's top tourism destination by arrivals yet ranks third in terms of tourism receipts while the USA is third in terms of international arrivals and ranks first in terms of tourism receipts. Suggest reasons for these differences.
- g** Match each of the countries listed in the World's Top 10 Destinations by Arrivals with their attractions listed below.

Country	Attractions
	Banff National Park, Vancouver Island, skiing in Whistler
	Disneyland, The Big Apple, cable cars of San Francisco
	La Rambla, Gaudi Architecture, Museo Guggenheim de Arte Contemporaneo
	Cairngorm Mountains, Big Ben St Paul's Cathedral
	Winter sports in the Alps, Vienna, Eisriesenwelt Caves
	Beaches of Acapulco, Spanish built city of Oaxaca, Mayan sites
	Rhine Valley, museums of Munich, Frankfurt
	The Forbidden City, The Summer Palace, The Bund
	The Seine, Cannes Film Festival, Mont Blanc
	The Vatican, Amalfi Coast, Florence

Tourism spenders

World's top 10 tourism spenders

Rank	Country	International tourism expenditure (US\$ billion)
1	USA	58.0
2	Germany	53.2
3	UK	40.4
4	Japan	26.7
5	France	19.5
6	Italy	16.9
7	China	15.4
8	Netherlands	12.9
9	Hong Kong (China)	12.4
10	Russian Federation	12.0
	WORLD	474.0

[Source: World Tourism Organisation, Tourism Highlights, Edition 2003]

- 3** Refer to the table above to complete the following:
- a** Locate the world's top 10 tourism spenders on a blank outline map of the world.
- b** Describe the spatial distribution of the world's top 10 tourism spenders.
- c** Explain the spatial distribution of the world's top tourism spenders.

Impacts of global tourism

4a Complete this table and list as many of the positive and negative impacts of global tourism as you can.

	Positive	Negative
Environmental		
Social		
Economic		
Political		

4b Discuss the statement that the benefits of global tourism far outweigh the negative impacts of global tourism.

The future of tourism

5 Construct a mind map that identifies possible threats to the growth of international tourism.

Note to teacher

Students will need to use an atlas and a blank outline map of the world to complete this worksheet.

Introduction The wine industry is a long-established industry that is an important part of the culture of many nations. While many traditions in the industry have remained it has experienced growth and expansion as it has spread from the old-world wine countries to the new.

Definitions

1 Match each term in the list below with the correct definition in the table.

fortified wine, terroir, verasion, vintage, viticulturalist, viticulture, wine-making, winery

1 The premises on which wine is made	
2 Someone who practices viticulture.	
3 The process of converting the sugar present in grapes into an alcoholic beverage.	
4 The science and practice of grape culture.	
5 The complex link between the soil and the taste of the wine.	
6 Wines that have had spirit added raising the alcohol level of the finished wine to between 17 and 20 per cent.	
7 Either the grape harvest or the wine produced in a single year.	
8 The period before harvesting when grapes change colour, start to soften and increase in size.	

Types of wine grapes

2 Classify the following grape varieties as either red or white.

Cabernet sauvignon, Chardonnay, Merlot, Pinot noir, Reisling, Semillion , Shiraz, Trebbiano

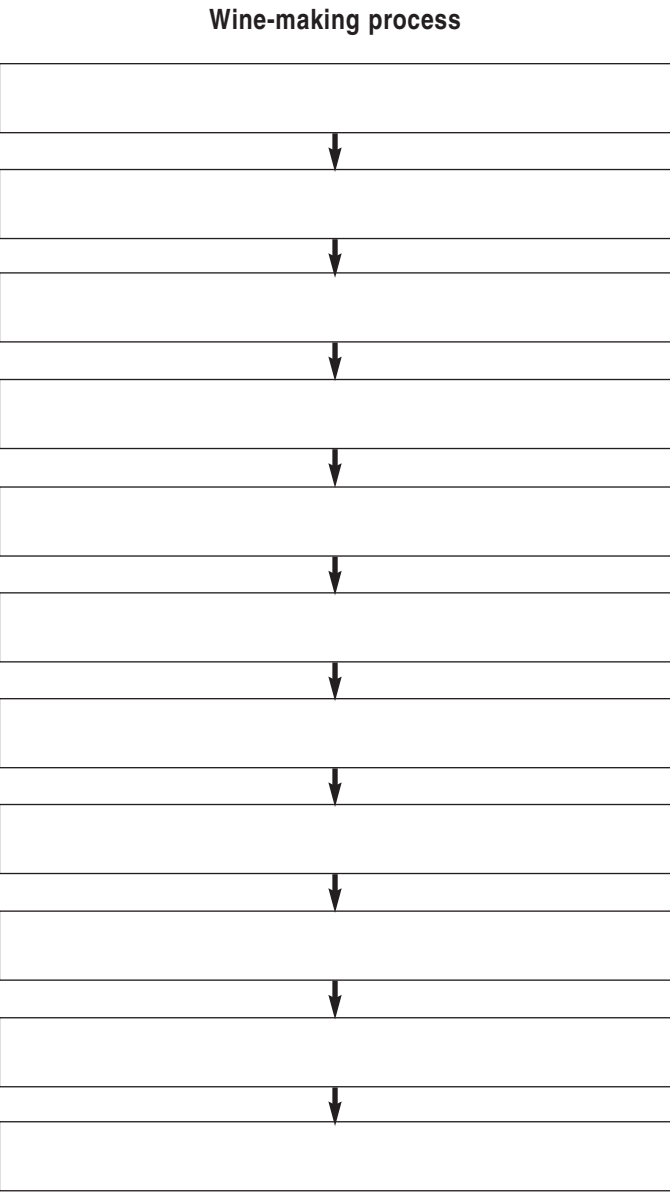
Red grape varieties	White grape varieties

The wine-making process

3a Place each of the following terms correctly in the flow diagram below.

bottling; clarification and stabilisation; crushing the mixture of juice, skin and seeds; distribution; fermentation; harvesting the grapes; maturation in the bottle; packaging; pressing; storage

3b In the last box, insert a term that best describes the general stage that follows the wine-making process.



Wine producers and consumers

Global top 10 producers of wine

		Wine production (ML)	
		1997	2001
1	France	5 356	5 824
2	Italy	5 089	5 130
3	Spain	3 322	3 113
4	USA	2 200	2 380
5	Argentina	1 350	1 215
6	Australia	617	1 077
7	Germany	849	966
8	South Africa	881	747
9	Portugal	612	701
10	Chile	455	600
	WORLD	26 050	27 544

[Source: Centre for International Economic Studies (CIES), University of Adelaide]

Global top 10 consumers of wine

		Per capita wine consumption (litres)	
		1997	2001
1	France	60.6	59.5
2	Italy	53.6	52.3
3	Portugal	52.5	50.2
4	Croatia	45.1	46.1
5	Switzerland	40.9	40.8
6	Spain	37.1	38.5
7	Argentina	40.3	34.0
8	Uruguay	30.7	31.9
9	Hungary	29.0	31.0
10	Romania	29.6	30.0

[Source: Centre for International Economic Studies (CIES), University of Adelaide]

- 4a On a blank outline map of the world label the top ten producers of wine globally.
- 4b On a second blank outline map of the world label the top ten consumers of wine globally.
- 4c Discuss whether there is a correlation between countries, which are producers of wine and countries, which are consumers of wine. You should use examples in your discussion.

Who am I?

- 5 Identify the countries from the top 10 producers and/or consumers of wine that are described below.
- a I am a southern hemisphere wine-producing nation. I feature in the list of top 10 producers of wine but not on the list of top 10 consumers of wine. Due to sanctions I was prevented from exporting wine in the 1980s but have now developed a significant export market.
- b I am the world's largest producer of wine as well as the world's largest consumer of wine.
- c I am a country located in the southern hemisphere. My wine production increased by 75 per cent between 1997 and 2001.
- d I am the most important producer of wine in South America. I am also a large consumer of wine. My wine growing regions are located in the west of the country at the foothills of the Andes but are spread over a wide range of latitudes.
- e Located in Europe, I am both a very large producer and a large consumer of wine. Wine grapes are cultivated throughout the peninsula from the Alps in the north to the islands that are located close to the coast of north Africa.

SWOT analysis

- A SWOT analysis is used by many organisations to help identify important aspects affecting their operations.
- 6 You are an advisor to a successful local wine producer who wants to expand. Develop a simple SWOT list to help start the process.

S	Strengths	
W	Weaknesses	
O	Opportunities	
T	Threats	

SELECTED ANSWERS

1 ENERGY RESOURCES

1 biomass RM, coal N, crude oil N, geothermal N, hydro power RM, marine currents RP, natural gas N, ocean thermal RP, peat N, shale oil N, solar RP, tidal RP, uranium N/RM, wave RP, wind RP, wood RM

2 **b** renewable 15%, non-renewable 85% (assuming all nuclear is classified as non renewable) **c** fuels sourced from recently living organisms – includes wood, other plant material, animal dung and in the past, whale oil **d** indigenous peoples, rural dwellers in developing countries – they face shortages of fuel, it's collection damages their environment and burning such fuels creates human health risks **e** these are the more modern, technologically advanced fuels such as solar, wind, wave, tidal, ocean thermal – they are relatively expensive and provide low energy output compared to coal and oil at present – will become more viable with improving technology and increasing oil prices

3 **a** world energy 329%, world population 122%, energy use 1950 – 37, 2000 – 71.5 **b** energy use growing much faster than population/ use per person has approximately doubled in last 50 yrs – more rapid growth expected in future with increasing economic development

2 THE OIL CRISIS

1 **a** 1900 TR 40% coal 57% oil 2% NG 1%; 1950 TR 20% coal 50% oil 20% NG 8% HE 2%; 2000 TR 10% coal 22% oil 36% NG 22% HE 3% Nuc 7% **b** now more diversified, decline in traditional fuels and coal, rise in nuclear and natural gas, oil has fluctuated **c** sharp drop in oil's share 1970–1980 with rapid rise in share from natural gas and nuclear – response to increasing price of oil **d** graph shows percentage shares, not totals used – total energy consumption has been rising **e** modern renewables such as solar, wind generation, tidal – still comprise a very small proportion of total energy supply **f** mostly biofuels such as timber, peat, dung

2 Sector sizes – crude oil 126°, coal 79°, natural gas 76°, nuclear 25°, hydro power 7°, biofuels 11°, renewables 7°

5 **a** North America 5.5%, Central and South America 8.9%, Europe and Eurasia 9.2%, Middle East 63.3%, Africa 8.9%, Asia Pacific 4.2% **b** 2044 (2003 + 41) **c** North America, Asia Pacific, Europe and Eurasia, Africa, Central and South America, Middle East

3 WATER RESOURCES

1 Vocabulary – terms in table order – water withdrawal, renewable water, aquifer, water stress, non-renewable water, water scarcity, sanitation.

2 (Ocean) **salt water** – abundant supply on coastlines, but limited use, mainly for industrial cooling – desalination is expensive and involves high energy consumption – high initial capital cost to pipe inland / plus pumping costs **glaciers and icecaps** – considerable global supply but mainly in remote and sparsely populated locations, water however is fresh – limited use for local consumption, expensive to relocate supply **groundwater** – some supplies available in arid areas (eg. inland Australia) but may be too saline for irrigation and human use without expensive treatment. Often described as 'fossil water' as recharge rates are much slower than human extraction – many bores running dry **surface water** – most widely used, enhanced by dam construction but subject to variable climatic conditions and easily polluted – in need of increasing levels of purification

3 **b** **USA** by far the highest, due to ample supply network and high living standards. Much is used in industry, agriculture is irrigated extensively in arid regions and domestic consumption levels are also high. **Namibia** has the lowest, reflecting an arid climate and poor supplies, low living standards, and little industry and limited irrigated agriculture. Little wealth is available for supply infrastructure. **Egypt and Australia** have similar levels, but more of Egypt's is extracted from the Nile for irrigation, while Australia uses a bigger share for

urban domestic purposes and industry. **France** is more typical of European patterns, where more humid climates and lower temperatures (except in the south) result in lower demand for irrigation. It uses a high proportion in its manufacturing industries. **China** has the second lowest per capita use, with the largest proportion used to irrigate in order to feed its massive population, but domestic and industrial use are growing with economic development.

4 **a** Agricultural 65%, Industrial 25%. Municipal 10% **b** In 1900, most used for agriculture and little for industry. Population growth, industrialisation and economic development, particularly post 1950 has seen total use, and particularly the proportion used for industry rise dramatically.

5 **a** Sector sizes for graph (in degrees)

	Domestic/public	Industrial	Agricultural
Egypt	22	29	309
USA	43	166	151
Australia	234	7	119
Germany	50	245	65

b See answer for previous question for country detail. Total withdrawals reflect availability, population size, standard of living and patterns of production.

6 Measures might include management of current supplies to increase volumes, including more desalination, improve quality, increase recycling, reduce waste, redistribute supplies or people. Population growth controls would assist in limiting the growth in demand.

4 GLOBAL PATTERNS IN FISHING

1 **a** The main widely used commercial sources include aquatics - fish, shell fish (eg. oysters) and crustaceans (eg. prawns) and terrestrials such as cattle, sheep, pigs, goats poultry, milk and milk derivatives such as cheese, and eggs. Specific regions and cultures also consume more specialised products such as kangaroo, dog, and whale meat. Indigenous communities consume these and a wider range of land mammals and bird protein, aquatic mammals such as seals and dugong, as well as reptiles and rodents. Small animals such as grubs, and insects at different stages of their lifecycle, are also an indigenous animal protein food source. **b** Total world supply – 95.5 million metric tonnes; Percentage shares – Africa 6.5, North and Central America 8.5, South America 3.0, China 32.7, Other Asia 34, Europe 14.6, Oceania 0.7 (rounded to 1 decimal place) **c** China consumes approximately one third of the world's catch and its share has been growing rapidly. If added to the rest of Asia, it would distort the regional figures and its impact would not be as apparent.

d Oceania is a region with a relatively small population, but many small island nations that, due to both limited suitable land area and tradition, rely heavily on seafood.

2 Sector graph segments – consumed fresh 193°, frozen 92.5°, canned 39.5°, cured 35°

3 **a** Advantages include the preservation and regeneration of natural fish stocks and eliminating wasteful by-catch. Fishing methods such as bottom trawling that damage habitats would also be reduced. There is also the potential for more regular supply to match demand and the lack of dependence on natural conditions such as weather and catch size variations. Disadvantages are that many farmed species need aquatic sourced food, which would have to be caught, pollution of ocean waters from coastal fish farms and the risk of 'crop failure' from disease and other production problems. **b** Inland fishing – from lakes and rivers eg. Canada's Great Lakes and Europe's Danube River. Marine fishing – from open oceans and coastlines. **c** Capture fluctuates, indicating the problems of hunting for an uncertain prey in variable seasons, while the steady growth in aquaculture reflects the controlled environment and agricultural/industrial methods of production.

4 Actions in table order of effects – wetland depletion, use of driftnets, bottom trawling, overfishing, water pollution, dam building on coastal rivers, catch quotas

5 TROPICAL RAINFORESTS AND DEFORESTATION

- 1 a biodiversity b infrastructure c ecosystem d leaching e tropical rainforest f watershed
- 2 a 1 False. Over 50% of the world's plant and animal species live in rainforests. 2 False. Brazil has the largest area of Tropical rainforest in the world. 3 True 4 False. When rainforest trees are cut down they release carbon dioxide into the atmosphere contributing to global warming. 5 True 6 True
- 3 b Pangkalanbun has high temperatures and heavy rainfall all year round. c 1°C d Due to its location so close to the equator the sun is almost directly above all year round. e Rainfall is heavy all year round, although heavier from November to May. f The rainfall would reduce due to reduced evapotranspiration from rainforest trees.

4 a

1	saprophytes
2	epiphytes
3	parasites
4	leaf litter
5	humus
6	lianas
7	herbivores
8	carnivores

5 a

Causes	Impacts
- cattle ranching and livestock grazing - commercial agriculture - fuelwood collection - infrastructure development - logging - mining and oil exploration - slash and burn farming - transmigration	- destruction of traditional lifestyle - destruction of waterways by high sedimentation loads - drought and desertification - global warming - loss of biodiversity - soil leaching - soil degradation

6 DESERTIFICATION

- 1 Vocabulary terms in table order – drought, overgrazing, ecological refugees, desertification, afforestation, evapotranspiration, salinity, biodiversity, fuelwood, windbreaks
- 2 a Area 20 008 (000 km²), Population 281.23 millions b Area ranking – Asia, Africa, North America, South America, Australia, Europe; Population ranking – Asia, Africa, South America, Europe, North America, Australia c The most logical ranking is Asia, Africa, South America Europe, North America and Australia if people affected and the ability of the region to deal with the problem are major criteria. d Sector sizes for the graph – Africa 133°, Asia 136°, Australia 20°, Europe 5°, North America 37°, South America 29°
- 3 a **Collecting fuelwood** denudes the landscape of vegetation, increasing both wind and water erosion, reduces the fertility of soil through lack of humus and increases evaporation through reduced shade **grazing /overgrazing** destroys ground cover from animal consumption, and leads to compaction of soil by hard hooved animals – erosion increases and vegetation is less able to regenerate when grazing ceases **drought and/or climate change** reduce the capacity for plant regeneration + the plants are less suited to the arid conditions and their habitat is made less inviting due to increased erosion, exposure, temperature changes and reduced soil reduced moisture content **overcultivation** does not allow soil to regenerate, so soil moisture and fertility decline, making future harvests less abundant or resulting in crop failure **population pressure/permanent settlement** lead to overcropping as agricultural land becomes scarce due to urban expansion and demand for food / distances travelled may increase from rural villages to farmland which is more marginal than that already in use **poverty** forces people to use marginal land and degrade their environment due to shortage of resources (land, fuel) and does not allow them to make

- improvements which raise productivity and protect their environment – poor landless tenant farmers have little ability or incentive to invest in land protection or improvement **inefficient irrigation** wastes the water resources available which may be in short supply, through runoff, evaporation or waterlogging – this may also result in salinity problems **government priorities** may skew investment away from agriculture and land remediation toward urban and industrial development, or as in the case of some African and Asian countries, funds may be diverted into political and military conflict **conflicts** both divert resources and increase degradation and people are forced to abandon their land / military vehicles and use of heavy arms destroy vegetation, damage water supply systems and increase erosion.
- 4 1 remote sensing (satellite imagery) can help determine which areas are at risk and resources can be more appropriately allocated 2 computer simulation (GIS) can evaluate the effects of changing climatic conditions, landuse practices and the likely success of remedial action 3 leguminous plants(used in crop rotation) add valuable nitrogen to the soil, thereby increasing fertility and increasing crop yields 4 stones help to stabilize newly planted trees, reduce wind stress on roots and help retain soil moisture lost to evaporation 5 rows of trees or shrubs provide wind breaks that reduce wind erosion and provide shelter for animals 6 spraying dunes with oil keeps them stable (still) until the seeds germinate and grow sufficiently to arrest dune migration 7 planting woody trees and shrubs close to villages provides a fuel source so natural vegetation is not destroyed and reduces time needed for fuel collection that can be diverted into better land management 8 imported food reduces the problems of overgrazing and allows natural vegetaion to regenerate, reducing desertification 9 solar cooking devices reduce the denudation of vegetation and associated problems 10 contour channels (or ploughing) help retain runoff after rain, thereby reducing water erosion and increasing soil water content 11 straw bale grids around new tree seedlings protect them from wind and reduce evaporation from watering – as the straw decomposes it enriches the soil – it provides a habitat for small organisms that may be directly or indirectly beneficial to the plant 12 greater crop variety can act as insurance against crop failure, and commercial crops can provide income for land improvement projects 13 these can provide income that is used to import food and reduce pressure on agricultural land 14 community projects such as tree planting and improved irrigation systems can result from use of surplus or under-employed labour, supported by aid funds 15 cropping produces more food per area than crops – this may reduce land areas utilized and problems associated with the impacts of grazing 16 crop rotation can be used to replenish fertility using legumes (nitrogen), resting crops can be ploughed in to add humus and companion planting can reduce the need for pesticides – low water demand crops can be introduced onto more arid zones 17 rural –urban migration if appropriate and well managed reduces population pressure in marginal agricultural areas and therefore the impacts of agriculture on desertification 18 efficient water storage and irrigation (using less water or expanding the area irrigated) results in improved land quality as more land can be vegetated and soil does not suffer from salinity/waterlogging problems – productivity increases help a community undertake landcare 19 local people are more likely to participate in projects if they are consulted – demonstration projects are an important step in the process 20 recreational vehicles are a major cause of land degradation in the USA by destroying vegetation, compacting soil and increasing erosion.
- 7 POPULATION CHANGE
- 1 Terms – 1 demography, 2 median age, 3 fertility, 4 natural increase, 5 density, 6 distribution, 7 net migration
- 2 Population decrease – deaths and emigration, Population increase – births and immigration. Answers dependent on examples used
- 3 a False, True, False, True, False b Similarity – both have falling rates of growth; Differences – rate of change greater in less

developed regions +less developed regions will experience population increase, while more developed regions will experience population decrease

8 GLOBAL POPULATION GROWTH

1 a percentage increase b fertility rate c percentage share
d spatial distribution e life expectancy f natural increase g infant mortality

2 a During the 19th century the world's population growth was much slower increasing from 1 million to 2 million. In the twentieth century the world's population grew much more rapidly increasing from 2 million to 6 million. b improved health and sanitation, better nutrition, improved medical services c It took 170 years for the world's population to grow from 1 million to 4 million while in 55 years from 1999 to 2054 the world's population is expected to increase from 6 million to 9 million. d Infant mortality rates have fallen and life expectancy has increased due to better health, nutrition and medical services.

3 a more developed – 19%; less developed – 81% b There are 1 202 000 people being added to developed countries each year and 81 792 000 people being added to developing countries each year. c The fertility rate in developed countries is 1.5 children per woman on average compared with 3.1 children per woman on average in developing countries. Women in developing countries tend to have more children to ensure at least some will survive, to have a number of children to help at home on the farm or the family business, to have children to look after them in their old age. Women in developed countries are often having less children because they are pursuing a career, having children is expensive. d In developed countries only 18 % of the population is aged under 15 while in developing countries 33% of the population are aged under 15. In developed countries 15 % of the population on average is over 65 compared with 5 % of the population in developing countries. In developed countries the biggest challenge is going to be catering for an ageing population with medical, hospital and nursing home facilities. In developing countries the biggest challenge will be providing a young population with education and employment. e There is a huge disparity in life expectancy between developed and developing countries because of variations in access to health and medical facilities and nutrition. f People in less developed countries tend to work in agriculture and often subsistence agriculture therefore they live in rural areas. People in developed countries work in a whole range of industries many in services and manufacturing. They buy agricultural products which are produced commercially.

4 a Spread across the globe although heavily concentrated in the Southern hemisphere – Middle Africa, Western Africa, Latin America and the Caribbean, South America and Western Asia. b Life expectancy has increased while fertility rates have remained relatively high. c Southern Africa and Europe are expected to experience decline, Eastern Asia and Northern Europe are expected to grow at a much slower rate. d The population of Southern Africa will decline because of the impact of the AIDS virus, the population of many countries in Europe is declining because women in these countries are having fewer or no children

5 a Russia and Japan. Developing countries with high population growth rates are moving into the top 10 largest countries in the world. b China – 26%, India – 30% c China has a One Child Policy whereas population control in India is not as strongly enforced. d Five of the largest countries in the world will be in Asia, three will be in Africa plus Brazil and the United States.

9 THE AGEING OF THE WORLD'S POPULATION

1 a fertility rate b life expectancy c developing countries
d developed countries **2** a 2.9 children b In developed countries children are seen as an expense, women in developed countries are leaving their child bearing till later and therefore having less children, women are pursuing a career, children in some developing countries now have a better survival rate so there is no need to have as many,

better access to contraception. c 32 years d better nutrition, better health and sanitation, better access to medical services

3 a 1 334 893 000 b Europe – 14%, North America – 8%, Oceania – 10%, Asia – 13%, Latin America/Caribbean – 12%, Near East/North Africa – 9%, Sub Saharan Africa – 2%

4 a The potential support ratio is the number of persons aged 15–64 years to each person aged over 65. b The psr is expected to fall from 9 people in the 15–64 year age group for every one person aged over 65 to 4 people in the 15–64 year age for every person aged over 65. c In both developed and developing nations there will be fewer people working to support those in retirement. In developed countries they have been preparing for this by encouraging people to put money away in the form of superannuation, also governments provide means tested pensions although this will become increasingly difficult as the ratio of people working to those in retirement decreases. Taxes will need to increase to pay for this. There will also be a greater need for medical facilities, hospitals and nursing homes. Developing countries will have similar needs. However they will not have the means to pay for them. In many of these countries families play a much greater part in looking after the elderly. d Currently those aged 80 plus make up 12 % of the population aged 60 years or older by 2050 they are expected to make up 19% of the population aged 60 years or older. e increase in life expectancy due to better health, better nutrition, improved medical facilities; decrease in fertility rates due to better access to contraception, preference particularly in developed countries for having fewer children as women pursue careers and leave child bearing until later

5 a False Increasing life expectancy and falling fertility rates both help to explain the ageing of the world's population. b True c True d False. Less developed countries will experience the greatest numerical increase in the number of people ages over 60 between 2002 and 2050. e False. The region with the highest percentage of its population over the age of 65 is Europe followed by North America and Oceania. f True

10 CHINA'S ONE CHILD POLICY

1 a 75% b 37% c The percentage increase in the second 30-year period dropped dramatically due to the impact of China's One Child Policy.

2 a Between 2000 and 2025 China's population is expected to age, with a much greater proportion of the population in the older age groups as people start to live longer and fewer people in the younger age groups as people in recent years have had less children and are continuing to do so. b China will now have many more people to cater for in retirement. There will be a need to provide them with more health and medical facilities.

11 THE GLOBAL AIDS CRISIS

1 a The global AIDS crisis is heavily concentrated in Africa. In 2001, 11 million of the 13 million children in Africa, Asia and Latin America who had lost a parent to AIDS lived in Africa. b Pandemic means prevalent throughout an entire country or continent, or the whole world and is often used to describe the spread of disease.

2 a 2.2 million people died of AIDS in Africa and the Middle East in 2001. b 2.75 million people died of AIDS globally in 2001.

c People who died of AIDS in Africa and the Middle East in 2001, comprised 80% of global AIDS deaths in 2001. d There are much greater numbers of people using antiretroviral drugs in high income countries than in the less well off countries of Africa and the Middle East, Asia and Eastern Europe in particular. This is due to the cost of these drugs.

3 a 26 million b 1300%

4 a 13 million b 24 million c 85% d Latin America and the Caribbean 500 000, Asia 1500 000, Africa 10 500 000

5 a True b False. In 2002 there were 42 million people living with AIDS worldwide. c False. The region with the highest incidence of

AIDS is Africa. **d** True **e** False. Antiretroviral drugs are more readily available in developed countries where profits can be made. **f** True

12 WORLD FOOD SUPPLIES

1 a Both are increasing, but world food production has been growing more rapidly than population, with a significant increase in food since 1980 – food has risen approx. 120% and population 80% over the period. **b** Increases in food production have not necessarily been directed to areas in need – there is a distribution problem with wealthy areas having abundant supplies and some countries, particularly in Africa, still experiencing chronic shortages. **c** Much of the world’s food is produced and consumed at subsistence level, with a lack of data recording. Different time periods for data collection and systems of measurement would also complicate the process.

2 a The people of the developed world consume more food than the developing world, and developing nations have less food variety and greater dependence on cereal crops. **b** Food and Agricultural Organization – part of the United Nations **c** Developed World – 1 cereals, 2 other, 3 vegetable oils, 4 sugar, 5 meat, 6 milk./dairy, 7 roots /tubers, 8 pulses. Developing World – 1 cereals, 2 other, 3 vegetable oils, 4 sugar, 5 meat, 6 roots/tubers, 7 milk/dairy, 8 pulses **d** Combine answers in **c** **e** Pattern of order is similar, but peoples of the developed world have more variety due to lower dependence on cereals such as rice, wheat and corn. They also consume about 55% of the total supply. **f** Developed world due to variety, but higher sugar intake could be a problem **g** Developing peoples consume much less meat, milk and dairy which would require refrigeration for storage – cereals and most other foods consumed in larger volumes are more easily stored. **h** Developed world – able to pay for transport and storage and source foods from other climates and overcome seasonal factors – have better developed infrastructure for wider food access. **i** Developing have greater dependence on crop food sources and those requiring less processing. Animal protein is expensive to produce in term of money, water land and capital. **j** These figures are averages for a world divided into two major groups. The average calorie intake may be adequate, but averages hide variations within the group – many people still die of starvation and suffer malnutrition.

3 a Flood – crops inundated and swept away, soil eroded, transport routes damaged, stock drowned **Refugees** – increase demand for food supplies often beyond local ability to provide **Drought** – reduces food output through crop failure, death of stock **Conflict** – diverts resources from food production and supply, may prevent supplies arriving, food used to feed fighters, add to refugee problem **Economic problems** – poor farmers forced to use seed stock and breeding animals for survival, unable to pay rent for land to farm, high land rents, high debt repayments **b** other natural hazards (earthquake, eruption, cyclone) pest and disease in crops and livestock, delivery failure through transport problems

4 a The Food and Agricultural Organization – a division of the United Nations **b** annually **c** To halve the number of hungry people between 1990 and 2015. **d** There is food available to solve the problem, but governments must create the mechanism for its effective allocation. **e** The 2004 Darfur conflict in Sudan illustrates this problem well – a large number of refugees, combined with reduced food production due to natural and human factors, civil war and violence and the inability of aid provided to get through to the needy. **f** Sector sizes – developing countries 341°, countries in transition 14°, developed countries 5° **g** Supporting statistics from passage include regional patterns and examples of individual countries. **h** Evidence for both – progress in some areas, decline in others/areas experiencing political conflict are most prone to hunger crises.

13 URBANISATION

1 Between 1970 and 2015 the urban population as a percentage of the total population in the world’s developing countries is expected to

increase by **24.6%** while the level of urbanisation in industrialised countries is expected to increase by **11.6%** The reason the increase in the proportion of people living in urban areas is much higher in **developing** than in **industrialised** countries is because the level of urbanisation was already high in **industrialised** countries in 1970. Least developed countries are also expected to experience a significant increase in urbanisation. However this increase will not be quite as **high** as for developing countries as a whole. The rate of growth of the urban population is expected to be **lower** in all regions of the world in 1995-2015 than it was in **1970–1995**.

2 a

Push factors	Pull factors
• Famine • Drought • Flood • Breakdown of the family farm through inheritance into unviable lots • Buying up of rural land by large companies. • Competition from large agribusiness.	• Employment • Bright lights of the city • Desire for an education for children. • Better health and medical facilities.

3 b In 2001, 11 of the world’s 17 megacities were located in Asia, 3 in South America, 2 in the USA, and Mexico City. **c** Sao Paulo, Mexico City, Mumbai, Calcutta, Dhaka, Delhi, Shanghai, Buenos Aires, Jakarta, Beijing, Rio de Janeiro, Karachi, Metro Manila **d** 13 of the 17 megacities were located in the developing world in 2001 – 76% **e** Sao Paulo – 78%, Shanghai – 12% **f** Sao Paulo – 3%, Shanghai – 0.46%

14 THE FORCED MOVEMENT OF PEOPLE GLOBALLY

1 b Definitions – 1 asylum seeker, 2 internally displaced person (IDP), 3 persons of concern 4 refugee, 5 UNHCR, 6 convention **2 b** The crisis in Rwanda **c** UNHCR has widened its mandate to include persons of concern and not just refugees because not all people who have been forced from their homes or experiencing political persecution are necessarily outside of their country. **5** 1 False, 2 False, 3 True, 4 True, 5 False, 6 False, 7 False

15 MEASURING DEVELOPMENT

2 a 1 adult literacy rates, 2 life expectancy, 3 purchasing power parity, 4 GDP, 5 Human Development Index, 6 GDP per capita, 7 Gini index, 8 under five mortality, 9 maternal mortality ratio

3a Life expectancy Low (under 65) - Democratic People’s Republic of Korea, Mongolia, Cambodia, Lao People’s Democratic Republic, Myanmar (Burma), Afghanistan, Bangladesh, Bhutan, India, Nepal, Pakistan; Medium (65 – 74) Indonesia, Malaysia, Philippines, Thailand, Vietnam, Iran, Sri Lanka; High (75 and over)- Hong Kong SAR, Japan, Republic of Korea, Singapore **Under 5 mortality rate male** Low (greater than 40) - Democratic People’s Republic of Korea, Mongolia, Cambodia, Indonesia, Lao People’s Democratic Republic, Myanmar (Burma), Vietnam, Afghanistan, Bangladesh, Bhutan, India, Nepal, Pakistan; Medium (11- 40) – China, Malaysia, Philippines, Thailand, Iran, Sri Lanka; High (10 or less) – Hong Kong SAR, Japan, Republic of Korea, Singapore **Under 5 mortality rate female** Low (greater than 40) – China, Democratic People’s Republic of Korea, Mongolia, Cambodia, Indonesia, Lao People’s Democratic Republic, Myanmar (Burma), Afghanistan, Bangladesh, Bhutan, India, Iran, Nepal, Pakistan; Medium (11 – 40) – Malaysia, Philippines, Thailand, Vietnam, Sri Lanka; High (10 or less) – Hong Kong SAR, Japan, Republic of Korea, Singapore **GDP per capita (PPP)** Low (Below US\$9000) – China, Mongolia, Cambodia, Indonesia, Lao People’s Democratic Republic, Myanmar (Burma), Philippines, Thailand, Vietnam, Bangladesh, Bhutan, India, Iran, Nepal, Pakistan, Sri Lanka; Medium (US\$9000 - US\$23 999) – Republic of Korea, Malaysia; High (US\$24 000 and above) – Hong Kong SAR, Japan, Singapore **Access to safe water** Low (less than 75%) - Mongolia, Cambodia, Lao People’s Democratic Republic, Myanmar (Burma), Afghanistan, Bhutan; Medium (75 – 94%) – China, Republic of Korea, Indonesia, Philippines, Thailand, Vietnam,

India, Iran, Nepal, Pakistan, Sri Lanka; High (95% and above) – Democratic People's Republic of Korea, Hong Kong SAR, Japan, Malaysia, Singapore, Bangladesh

3b In many cases there is a correlation eg: The Democratic People's Republic of Korea, Mongolia, Cambodia, Lao People's Democratic Republic, Afghanistan, Bhutan have been categorized as having low development using all indicators.

Hong Kong SAR, Japan and Singapore are categorized as having high development using all indicators.

In others there are exceptions eg: Malaysia has been categorised as having medium development using all indicators except safe water where it is categorised as having high development.

For some indicators Indonesia has been categorized as having medium development for others low.

16 INTERNATIONAL AID

1 1 bilateral aid, 2 multilateral aid, 3 emergency aid, 4 project aid, 5 non government organisation, 6 official development assistance, 7 untied aid, 8 tied aid

3 a United States, Japan, France **b** United States-22nd, Japan-19th, France – 7th **c** 5 **d** Norway, Denmark, Netherlands, Luxembourg, Sweden

17 DIVERSITY IN ASIA

1 b China, Japan, Nepal, Taiwan, Vietnam, India, Laos, Cambodia, Indonesia, East Timor **c** Climate varies according to latitude. The sun is directly above the equator at 0°. Therefore the cities located in the low latitudes either south or north of the equator tend to be warmer than those further away from the equator. **1c** The closer a city is to the equator the warmer its average temperature. This is due to the sun being directly above the equator. This association is evident from the data in the table which shows Dili which is located 8.34S as having an average temperature of 25°C whereas Beijing which is located 39.56N has an average temperature of 12°C. Altitude can also affect climate. Temperature falls 6°C every 1000 metres above sea level. This is due to the earth being heated directly by the sun.

2 a Indonesia, China, East Timor, Philippines

3 China – 1.3 billion, Japan – 127.8 million, Singapore – 4.3 million, Vietnam – 82.5 million, India – 1.08 billion, Sri Lanka – 19.2 million

4 Mandarin – China, Tagalog – Philippines, Sinhalese – Sri Lanka, Tetun – East Timor, Lao – Laos, Kachen – Burma (Myanmar)

5 a Indonesia – Netherlands, Philippines – Spain, Vietnam – France, Malaysia – Britain, East Timor – Portugal

6 a GDP per capita is the total value of all the goods and services produced in a country in a year divided by the population. PPP adjusts the figure to take into account the cost of living in that country. It gives us an indication of the total value of goods and services being produced in a country in a year divided by the population. However, it is an average figure income may not be spread evenly and it does not take into account subsistence agriculture. **b** Rank from top – Japan, Hong Kong SAR, Singapore, Republic of Korea, Malaysia, Thailand, China, Philippines, Sri Lanka, India, Viet Nam, Cambodia, Bhutan, Pakistan, Mongolia, Bangladesh, Nepal, Myanmar

18 CULTURAL INTEGRATION

Food	Country of Origin
dolmathes	Greece
gado gado	Indonesia
Fried rice	China
kaftah	Turkey
kebab	Lebanon
laksa	Malaysia
naan bread	India
pasta	Italy
sushi	Japan
tom yum soup	Thailand

3b There is a high concentration of French speaking countries in Africa, where there were a number of French colonies. There are also other French speaking countries in Europe and Vietnam in Asia and Vanuatu and New Caledonia in the Pacific.

3c French language could have impacted on the culture of the countries listed in a whole range of ways eg the history and traditions, food, sport, music, television shows.

Introduction of invention
1875 Telephone
1903 aeroplane
1930s Television
1974 Personal computer
1983 Internet

19 GLOBAL TOURISM

1 a 675 million **b** 2700% **c** rising disposable incomes, commercial jets, cheap oil, low promotional airfares after WWII

2 b The top 10 tourist destinations are heavily concentrated in Europe, with 7 of the top 10 countries in Europe. **c** Many of the tourists come from the wealthy nations of Europe and they travel to other European nations which are nearby. **e** France-11%, Spain – 7%, USA-6% **f** A major reason that France is the most visited nation in the world is because of its location in Europe near many other wealthy European nations. People can easily make short visits to France. The USA is further away from other wealthy developed nations. Therefore if people do visit the USA they tend to stay longer and spend more money.

Canada	Banff National Park, Vancouver Island, skiing in Whistler
USA	Disneyland, The Big Apple, cable cars of San Francisco
Spain	La Rambla, Gaudi Architecture, Museo Guggenheim de Arte Contemporaneo
Britain	Cairngorm Mountains, Big Ben St Paul's Cathedral
Austria	Winter sports in the Alps, Vienna, Eisriesenwelt Caves
Mexico	Beaches of Acapulco, Spanish built city of Oaxaca, Mayan sites
Germany	Rhine Valley, museums of Munich, Frankfurt
China	The Forbidden City, The Summer Palace, The Bund
France	The Seine, Cannes Film Festival, Mont Blanc
Italy	The Vatican, Amalfi Coast, Florence

3 b The top 10 tourism spenders are wealthy nations located in the northern hemisphere. A number of them are in Europe. **c** 6 of the top 10 tourism spenders come from wealthy European nations.

20 THE WINE INDUSTRY

1 1 winery, 2 viticulturalist, 3 wine-making, 4 viticulture, 5 terroir, 6 fortified wine, 7 vintage, 8 verasion

2

Red grape varieties	White grape varieties
Cabernet sauvignon	Chardonnay
Merlot	Reisling
Pinot noir	Semillion
Shiraz	Trebbiano

3 a harvesting the grapes; crushing the mixture of juice, skin and seeds; fermentation; pressing; clarification and stabilisation; storage; bottling; maturation in bottle; packaging; distribution **b** consumption

5 a South Africa **b** France **c** Australia **d** Chile **e** Italy

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